Article

Compact deep neural network models of the visual cortex

https://doi.org/10.1038/s41586-026-10150-1

Received: 22 November 2023

Accepted: 15 January 2026

Published online: 25 February 2026

Check for updates

Benjamin R. Cowley[1,2]✉, Patricia L. Stan[3,4,5], Jonathan W. Pillow[2,6] & Matthew A. Smith[3,4,5,6]✉

A powerful approach to understand the computations carried out by the visual cortex is to build models that predict neural responses to any arbitrary image. Deep neural networks (DNNs) have emerged as the leading predictive models[1,2], yet their underlying computations remain buried beneath millions of parameters. Here we challenge the need for models at this scale by seeking predictive and parsimonious DNN models of the primate visual cortex. We first built a highly predictive DNN model of neural responses in macaque visual area V4 by alternating data collection and model training in adaptive closed-loop experiments. We then compressed this large, black-box DNN model, which comprised 60 million parameters, to identify compact models with 5,000 times fewer parameters yet comparable accuracy. This dramatic compression enabled us to investigate the inner workings of the compact models. We discovered a salient computational motif: compact models share similar filters in early processing, but individual models then specialize their feature selectivity by 'consolidating' this shared high-dimensional representation in distinct ways. We examined this consolidation step in a dot-detecting model neuron, revealing a computational mechanism that leads to a testable circuit hypothesis for dot-selective V4 neurons. Beyond V4, we found strong model compression for macaque visual areas V1 and IT (inferior temporal cortex), revealing a general computational principle of the visual cortex. Overall, our work challenges the notion that large DNNs are necessary to predict individual neurons and establishes a modelling framework that balances prediction and parsimony.

One of the most influential computational models of sensory processing is the simple yet effective linear–nonlinear model, which uses a single spatiotemporal filter to describe the stimulus selectivity of a neuron[3]. Linear–nonlinear models largely explain responses of neurons in the retina and primary visual cortex (V1), and their filter parameters are readily interpretable[4–6]. However, linear–nonlinear models are not able to capture all properties of V1 neurons[7,8] and entirely fail to predict responses from neurons in the higher-order visual cortex, such as V4 and IT[9,10]. This makes it clear that more complicated models, such as stacking layers of linear–nonlinear models, are needed[11]. Indeed, a core observation in machine learning is that by substantially scaling model and dataset size, better prediction is all but guaranteed[12]: the current leading DNN models for computer vision tasks have more than 100 million parameters[13], whereas large language models boast billions of parameters[14]. Recent work has shown that large vision DNN models have internal representations predictive of V4 and IT responses both in human and in non-human primates[1,2,15]. The sheer size of these models—tens of millions of parameters—makes it next to impossible to understand the step-by-step computations between image and response. In this work, we apply Occam's Razor by seeking parsimonious models (that is, those with as few parameters as necessary) that still rival the large 'black box' DNN models in predicting neural responses. We hypothesize that these large DNN models comprise many filters that are either redundant, spurious or irrelevant for predicting visual cortical neurons, and removing these filters will yield more explainable models. To test this hypothesis, we leveraged recent advances in model compression techniques[16,17] to unearth parsimonious yet powerful models of V4 responses to natural images; we then dissected these models to better understand circuit-level computations of V4 neurons.

Compact models of V4 neurons

To identify the computations of a V4 neuron, we first desired a model—regardless of its size—that accurately predicted the response of the neuron to any natural image; we then sought to compress this model as much as possible to reveal its computations. Ideally, we would adopt a data-driven approach that directly trains a DNN model on the responses of a V4 neuron to many images[18], but recording from the same neuron over long periods of time is experimentally challenging. Instead, large task-driven DNN models have satisfactorily predicted V4 responses with fewer than 1,000 training images[10,15]. We reasoned that combining both the data-driven approach (with its high degree of expressiveness) and

[1]Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA. [2]Princeton Neuroscience Institute, Princeton University, Princeton, NJ, USA. [3]Neuroscience Institute, Carnegie Mellon University, Pittsburgh, PA, USA. [4]Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA. [5]Center for the Neural Basis of Cognition, Pittsburgh, PA, USA. [6]These authors contributed equally: Jonathan W. Pillow, Matthew A. Smith. ✉e-mail: cowley@cshl.edu; mattsmith@cmu.edu

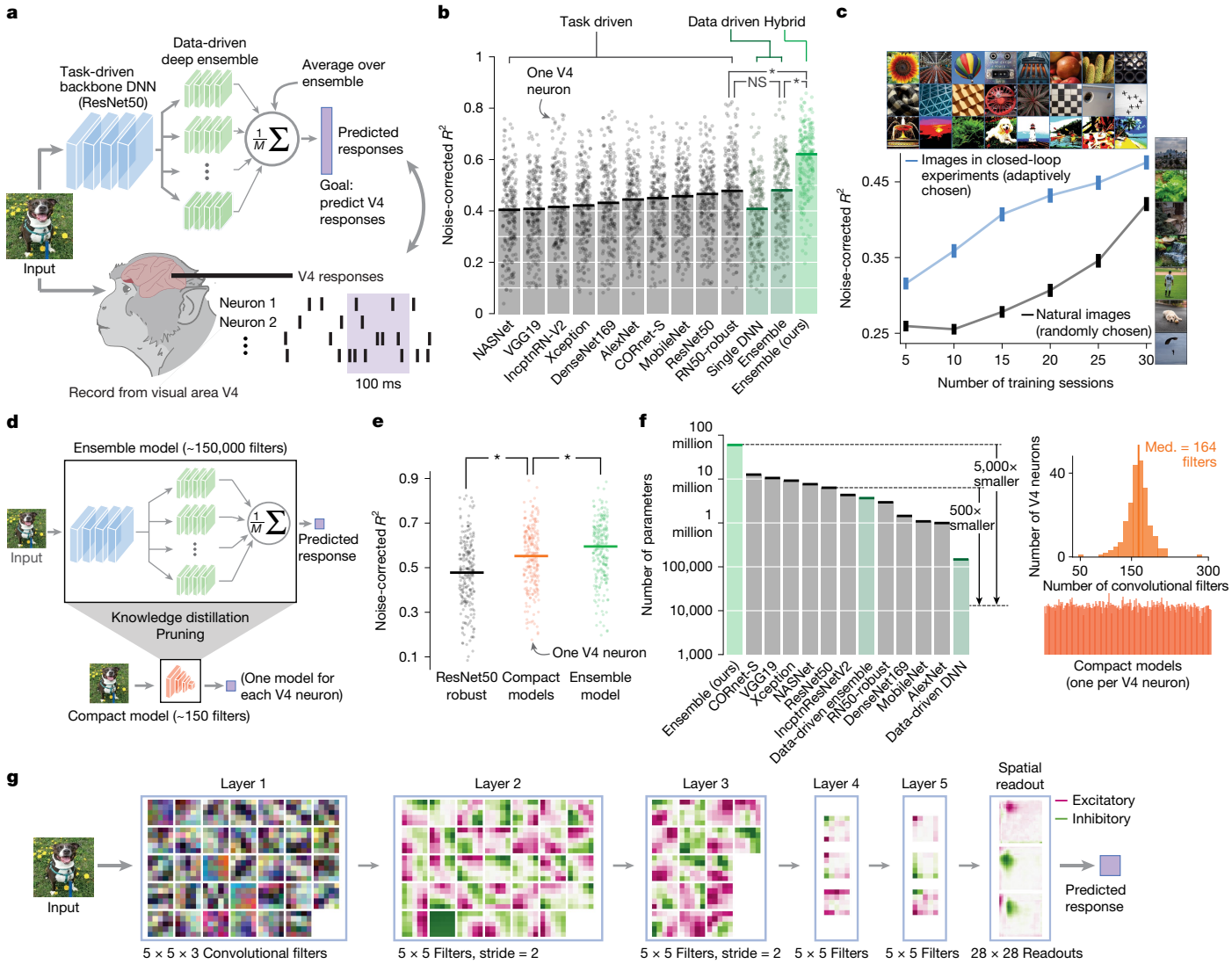


Fig. 1 | Identifying compact models of macaque V4 neurons. a, We sought to accurately model the mapping between natural images and repeat-averaged responses (spike counts taken in 100-ms bins) of macaque V4 neurons. Our hybrid deep ensemble model first extracts image features from a middle layer of a task-driven DNN (ResNet50) as a backbone (blue), and then passes these features as input into an ensemble of data-driven convolutional DNNs (green; see Supplementary Fig. 1 for detailed diagrams). The responses averaged across ensemble members are the final prediction. We thank S. Pattisapu for the macaque illustration. The macaque brain by M. Smith Breault was adapted from ref. 55, CC BY 4.0. **b**, Prediction performance for task-driven (black), data-driven (dark green) and our hybrid deep ensemble (green) models, evaluated on four held-out recording sessions from three animals. We report noise-corrected R^2, accounting for repeat-to-repeat noise. Each dot denotes one V4 neuron, and the lines denote medians. The asterisks denote significance (*$P < 0.001$, paired permutation test, $n = 219$ neurons); not significant (NS; $P = 0.890$). **c**, Prediction performance on the same held-out data in panel **b** when training on images adaptively chosen by the model during closed-loop experiments and gaudy images versus randomly chosen images (the insets show example images). See Methods for details. The lines denote medians, and the error bars denote 90% bootstrapped confidence intervals; $n = 219$ neurons. The images were reproduced from Adobe Stock. **d**, Model compression framework that uses knowledge distillation and pruning to compress our large deep ensemble ('teacher model'; top) to a compact model ('student model'; bottom), one for each V4 neuron. **e**, Prediction performance for the compact models (orange) versus ResNet50-robust (black) and the deep ensemble (green). Each dot denotes one V4 neuron and the lines indicate medians; *$P < 0.001$, paired permutation test; $n = 219$ neurons. **f**, Number of parameters for our deep ensemble (green), task-driven (black) and data-driven (dark green) models, and compact models (orange). The inset shows the total number of convolutional filters for each compact model; orange line denotes the median (Med.). **g**, Diagram of a single compact model showing all of its convolutional filters. Layer 1 filters are coloured based on their RGB channel input.

the task-driven approach (with its rich features of higher-order natural image statistics) into a hybrid approach offers even better prediction. To this end, we built a large hybrid DNN model that feeds an input image through a task-driven backbone DNN to obtain activity features from an intermediate layer (Fig. 1a, blue DNN); we then passed these features as input into an ensemble of convolutional DNNs (Fig. 1a, green DNNs). We used a deep ensemble for its ability to avoid overfitting in DNNs as well as its ensemble disagreement that serves as a measure of prediction uncertainty (see Methods). The parameters of the backbone DNN remained fixed, whereas the parameters of the deep ensemble were trained in closed-loop experiments in which the model selected the images of the next session with the largest ensemble disagreement to best refine its parameters. Our training data comprised 44 recording sessions from 3 monkeys; each session had approximately 50 neural units (a mix of single and multi-units recorded simultaneously with a chronic multi-electrode array; we refer to these units collectively as neurons) and approximately 2,000 unique images, leading to a combined total of over 78,000 unique images (see Methods). We held out

four recording sessions (at least one session from each of three animals) with new natural images for evaluation. Because we collected responses with chronically implanted electrode arrays, only a subset of neurons were likely to be the same across sessions. To train on as many neurons as possible, we gave each neuron in each session its own linear readout (Supplementary Fig. 1). This dataset distinguishes itself from previous benchmark datasets[15] with enough training images for data-driven DNNs to compete with task-driven DNNs[18,19].

We evaluated a suite of task-driven DNN models, trained to perform object recognition, on our V4 recordings and found prediction performances (Fig. 1b, black) to be consistent with those previously reported[15] (but see ref. 20 and Extended Data Fig. 2). We also trained a data-driven DNN (five-layer convolutional network) as well as an ensemble of data-driven DNNs (see Methods), the latter reaching prediction performance on par with the best task-driven DNN (Fig. 1b, RN50-robust versus data-driven ensemble). This highlights that our V4 dataset has enough recording sessions to serve as a benchmark for both task-driven and data-driven DNNs. The prediction performance of the hybrid deep ensemble model (Fig. 1b, green; noise-corrected $R^2 = 0.62$) was 30% better than the performance of the leading task-driven DNN and data-driven DNN model (noise-corrected $R^2 = 0.48$; Fig. 1b). This suggests that task-driven features provide a good prior for aligning to V4 tuning, but that refining these features with response data is key for better alignment. Here we focus solely on accurate prediction of our recorded V4 neurons without regard to generalizing to other V4 neurons from other datasets. Still, our hybrid deep ensemble, trained on our V4 responses, performs no worse than its backbone task-driven DNN when predicting V4 responses from a separate study[20] (Extended Data Fig. 2). These results suggest that a way forward in building a foundation model predictive of any visual cortical neuron is to combine the generalizable priors of task-driven DNNs with the specific features obtained from data-driven approaches.

To achieve this 30% boost in performance, we trained the deep ensemble model with closed-loop experiments via active learning[21,22]. This approach lets the model adaptively choose the images to show in the upcoming session that will best refine its parameters. The model chooses images about which it is most uncertain in its predictions, measured as the ensemble disagreement[23]. We also considered synthetically generated 'gaudy images' shown previously to improve training of DNN models[24]. Training on responses to model-chosen and gaudy images led to better prediction than that of randomly chosen natural images (Fig. 1c, blue line above the black line). To our knowledge, this is the first large-scale experiment in systems neuroscience to train a DNN model in closed loop. Other reasons for the 30% boost in prediction were the nonlinearities in the ensemble DNNs, training on enough recording sessions and enough ensemble members (Extended Data Fig. 1). The high prediction performance of our deep ensemble model suggested that this model and V4 neurons may share similar computations, but finding these computations embedded in the 60 million parameters of the model was next to impossible, a key hurdle in making DNN models accessible for human explanation.

We suspected that many of these model parameters were spurious or redundant[25]. Thus, we sought the most parsimonious model of V4 neurons by compressing the deep ensemble model as much as possible (that is, removing redundant and spurious filters) while maintaining high prediction power. Because too much compression may lead to a less human-interpretable representation—one cannot easily infer an object in an image from its JPEG compressed values—we constrained our compression procedure to identify shallow, feedforward neural networks (akin to a cascade of linear–nonlinear filter models[6]) with as few filters as possible. To perform model compression, we used the machine learning techniques of knowledge distillation, in which a smaller student model is trained to approximate a larger teacher model[16], and pruning, in which spurious filters are iteratively removed[17,26] (Fig. 1d). In concurrent studies, distillation was shown to slightly improve prediction performance of V4 responses but was not used for model compression[27]; V1 neurons[28] and fMRI voxel[29] activity were shown to be explained by simplified and compressed models. We first performed distillation by treating the deep ensemble as a teacher model and collecting its predicted responses to 12 million natural images. We then used these image–response pairs to train a much smaller DNN with 5 layers (100 convolutional filters per layer). Training this small DNN directly on our V4 data failed to outperform task-driven DNNs (Fig. 1b), hence the need for distillation. We then pruned this distilled DNN by ablating convolutional filters that contributed little to the output of the model followed by retraining; we stopped pruning when prediction performance decreased by 5% relative to that of the deep ensemble model. We called the resulting pruned model a compact model, one for each V4 neuron (Fig. 1d); having a single compact model for each V4 neuron (versus shared across V4 neurons, which we investigate later) ensures that all computations within a compact model solely contribute to predicting its matched V4 neuron. The compact models had a prediction performance (Fig. 1e; noise-corrected $R^2 = 0.55$) less than that of the deep ensemble (noise-corrected $R^2 = 0.60$) but greater than that of the top-performing task-driven DNN, ResNet50-robust (noise-corrected $R^2 = 0.48$). The compact models had approximately 5,000 times fewer parameters than the deep ensemble model (Fig. 1f, 'ensemble'), 500 times fewer parameters than its backbone DNN (Fig. 1f, ResNet50) and 250 times smaller than ResNet50-robust (Fig. 1f, RN50-robust). Indeed, a compact model had few enough convolutional filters (Fig. 1f, inset) for all of them to be viewed in a single diagram (Fig. 1g), similar to how the filter weights of linear–nonlinear models are inspected[6]. This result indicates that a substantial number of computations within large DNN models are either spurious or redundant when predicting the response of a single V4 neuron. The resulting compact models are small enough (close to 150 filters and 10,000 parameters for each model) to systematically interrogate their inner workings, which we pursue in later sections.

Preferred features of compact models

A natural first step in understanding the computations of the compact models is to identify their preferred stimuli. To this end, we synthesized an image that maximized the output of a compact model using gradient techniques (see Methods). Examples of these response-maximizing synthesized images (that is, preferred stimuli) revealed edges, curves, dots and textures (Fig. 2a), as expected from previous studies[19,20,24,30,31]. One concern about drawing conclusions from these preferred stimuli, which resemble but are not natural images, was that we had not experimentally validated in vivo whether these generated stimuli actually more strongly drove the responses of real V4 neurons relative to natural images[20,30]. To address this concern, we designed a set of experimental validations that followed the same procedure: we first trained compact models on previous recording sessions; we then probed each model for its preferred stimulus; last, we presented these probed images in a subsequent session along with randomly chosen natural images for reference.

We first probed each compact model for its 'preferred stimuli' selected from our large image database (Fig. 2b, left). These response-maximizing natural images strongly drove V4 responses above responses to randomly chosen images (Fig. 2b, orange dots above the black dots). We next probed the compact models by synthesizing images via gradient techniques to maximize the response of a given model neuron (Fig. 2c, left). We found that the synthesized maximizing images also elicited V4 responses well-above responses to randomly chosen images (Fig. 2c, red dots above the black dots). Our synthesized maximizing images may appear less crisp than the 'most exciting images' from other studies that leverage large task-driven and generative DNNs as image priors[19,20,31] partly because we are, to our knowledge, the first to optimize colour images, albeit with lower pixel resolution

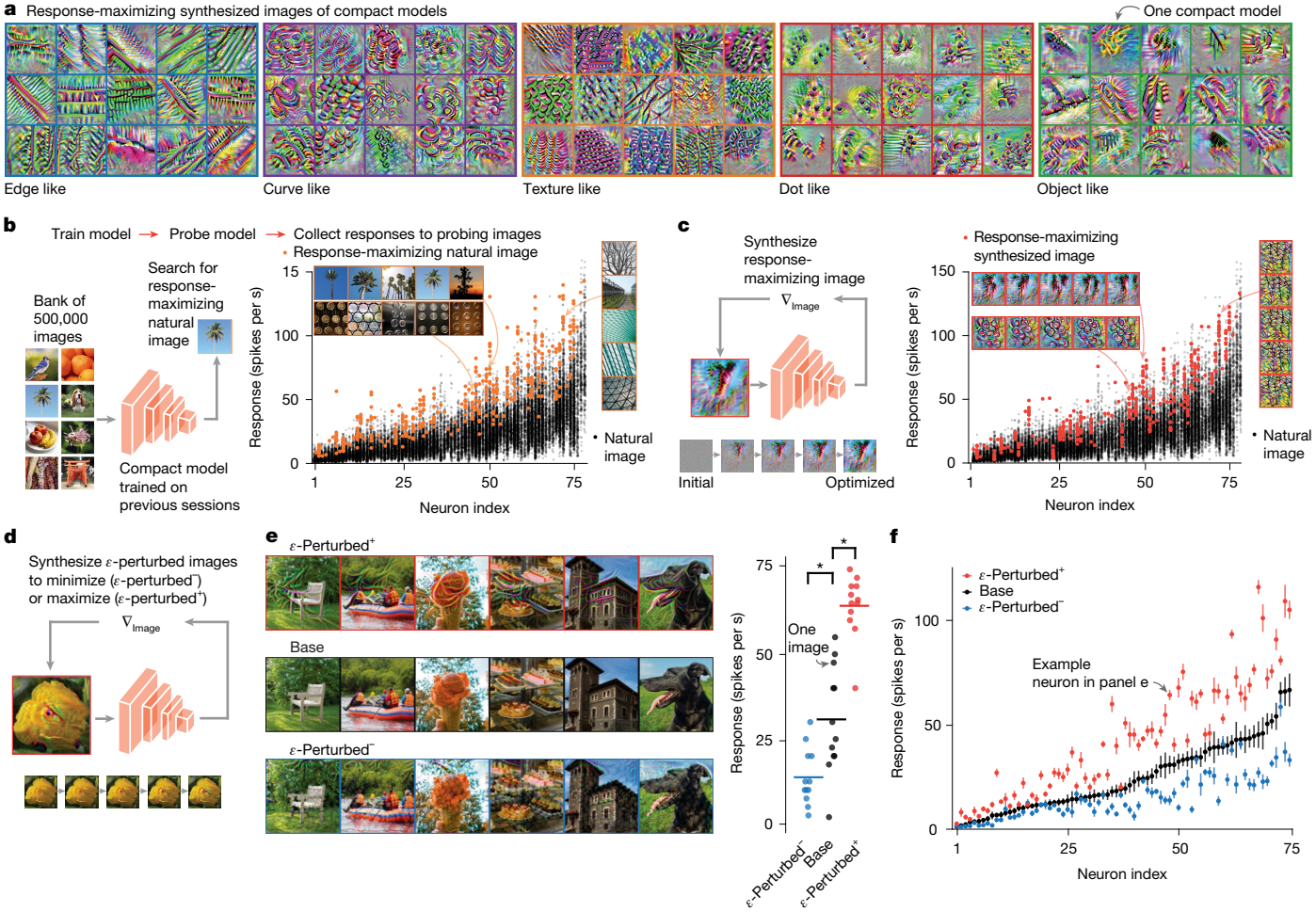


Fig. 2 | Experimentally validating the stimulus preferences of compact models. a, Example preferred stimuli of compact models (one stimulus per model; see Extended Data Fig. 3 for all compact models) optimized via gradient ascent to maximize the response of a model (see Methods). Stimulus categories are for illustrative purposes. **b**, Experimental validation in which we trained a compact model on previous recording sessions, probed the response of the model to 500,000 candidate natural images (left), and then recorded V4 responses to the natural images that elicited the largest predicted responses (right). The median response to these maximizing images (orange dots) was larger than 98.3% of responses to randomly chosen images (black dots; median across neurons, confidence interval 97.2–98.9%). Each dot denotes the repeat-averaged response to one image. The insets show response-maximizing natural images for three example neurons. **c**, Experimental validation for response-maximizing synthesized images predicted by the compact models. Each synthesized image started as a white noise image and iteratively changed via gradient ascent (bottom left inset). The median response to these maximizing images (red dots) was larger than 97.2% of responses to randomly chosen images (black dots; median across neurons, confidence interval 95.8–98.7%). The insets show example synthesized images (same neurons as in panel **b**). **d**, Experimental validation for ε-perturbed images predicted by the compact models. We optimized via gradient techniques each ε-perturbed image to either minimize (ε-perturbed$^-$) or maximize (ε-perturbed$^+$) the output of the model, stopping when the change in pixel intensities from the base image passed threshold ε (that is, a slight perturbation to the base image). **e**, ε-Perturbed$^+$, base and ε-perturbed$^-$ images for an example compact model (left). A real V4 neuron responded less to the ε-perturbed$^-$ images (blue dots; $P = 0.004$, paired two-sided permutation test) and more to the ε-perturbed$^+$ images (red dots; $P < 0.001$, paired two-sided permutation test) than responses to the base images (black dots). Each dot denotes the repeat-averaged response to one image; lines denote means; $^*P < 0.005$ ($n = 12$ images). Images in panels **b**,**d**,**e** were reproduced from Adobe Stock. **f**, ε-Perturbed$^+$ images drove larger responses across neurons (red dots; 83.6 ± 5.9% increase in mean response, $n = 75$ neurons), and ε-perturbed$^-$ images drove smaller responses across neurons (blue dots; 33.2 ± 2.5% decrease in mean response, $n = 75$ neurons) than responses to base images (black dots). The dots denote means, and the error bars denote 1 s.e.m. across $n = 10$–12 base images per neuron.

(see Methods; Extended Data Fig. 3). Still, it is reassuring that simple five-layer compact models are enough to strongly drive V4 responses without needing the priors of task-driven DNN models that may embellish their maximizing images (for example, synthesizing elaborate eyes for a V4 neuron that prefers two dots). We further confirmed that the maximizing images of the compact models targeted specific neurons rather than driving responses of the entire V4 population (Extended Data Fig. 4), and that for each neuron, the natural images with the largest responses (Fig. 2b,c, highest black dots) had similar image statistics to the maximizing images (Extended Data Fig. 4). Finally, we found that the response-maximizing natural images tended to drive V4 responses as effectively as the response-maximizing synthesized images (Extended Data Fig. 5). This suggests that one should not rule out searching over a large number of natural images (in this case, 500,000 images) when identifying the preferred stimulus of a neuron.

We viewed the experimental validation of response-maximizing natural and synthesized images as coarse tests: we strongly drove neurons with images outside the range of their usual working regime. We wondered whether predictions of the compact models also held while not deviating too far from the distribution of natural images. To this end, we slightly perturbed a natural image to yield large changes in the output of the model. In a similar manner to identify adversarial[32,33] or eigen-distorted images[34], we started with a base image and used a gradient technique to either maximize (ε-perturbed$^+$) or minimize

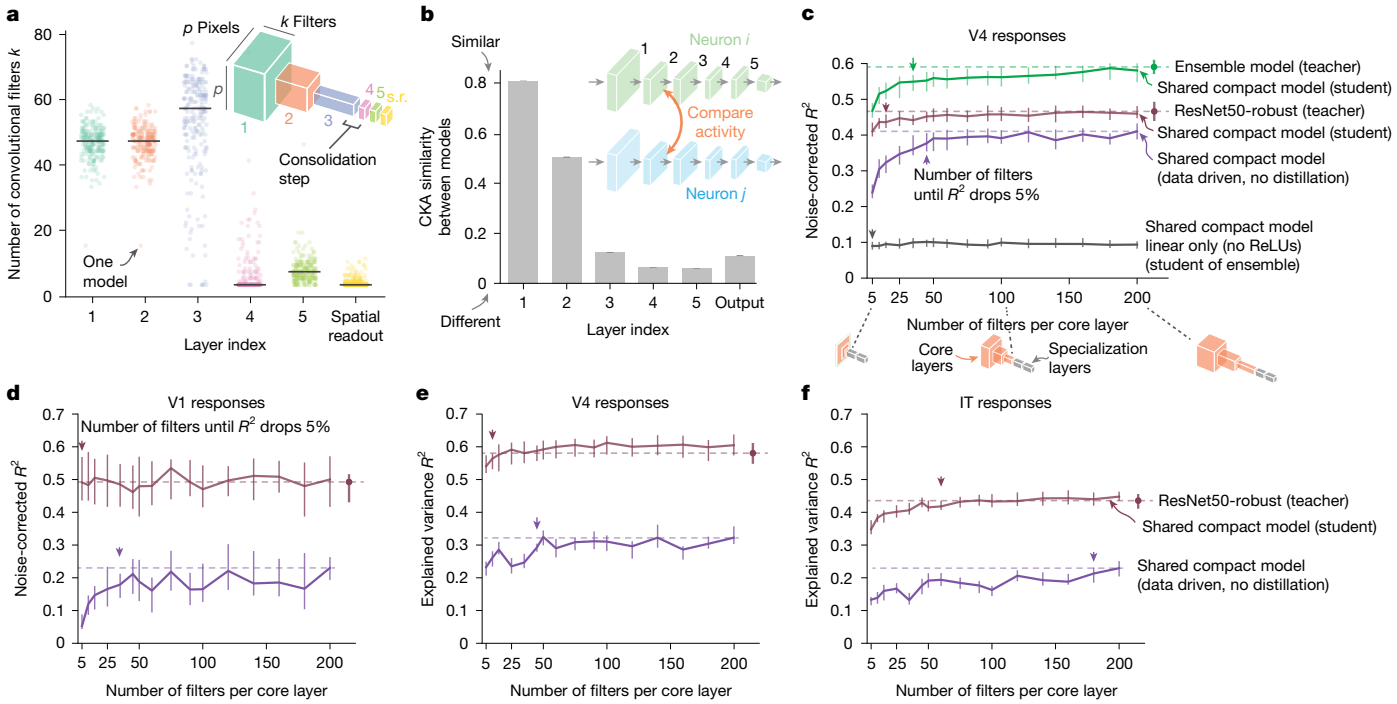


Fig. 3 | Compact models specialize their feature selectivity via a consolidation step. a, Number of convolutional filters per layer as determined by pruning; each layer had a minimum of 3 and a maximum of 100 filters. Each dot denotes one of 219 compact models; the lines denote medians. Layers 1 and 2 have the same number of filters as defined by the architecture of the compact model. The inset shows the diagram depicting shapes of activity maps ($p \times p \times k$ for p pixels and k filters) for each layer and spatial readout (s.r.). **b**, CKA similarity between the activity of the same layer for two compact models. The output activity of layers 1 and 2 had significantly higher CKA similarity than that of layers 3 and 4 ($P < 0.001$, paired two-sided permutation test, difference of means). The bars denote means; the error bars denoting 1 s.e.m. are small due to the large number of pairs of models ($n > 23{,}000$ pairs). **c**, Prediction performance of 'shared' compact models that predict all V4 neurons together. The number of filters in the first three 'core' layers was varied, whereas the remaining 'specialization' layers had 100 filters each (bottom diagrams). Each shared compact model was trained either via distillation as a student model with our deep ensemble (green) or task-driven ResNet50-robust (brown) serving as teacher models (dashed lines) or directly on real data (data driven, purple). We also distilled shared compact models with linear activation functions (black, 'linear only'). Traces denote medians; the error bars denote 90% confidence intervals, $n = 219$ neurons. The arrows denote the smallest number of filters with good compression performance, where R^2 remains within 5% of the R^2 for 200 filters. **d**–**f**, Model compression on neurons from different visual cortical areas spanning the primate visual system, using response datasets from macaque V1 (**d**)[41], as well as V4 (**e**) and IT (**f**)[40]. As expected, the data-driven compact models (purple) performed worse than the compact models distilled from a task-driven DNN (brown), as the amount of training data was small (less than 3,000 images) for each dataset. Same training procedures and format as in panel **c**; $n = 115$, 88 and 168 neurons for **d**–**f**, respectively.

(ε-perturbed$^-$) the output of the model (Fig. 2d). Once the perturbed image deviated too far from the base image (average difference in pixel intensity $\varepsilon = 10$; see Methods), synthesis stopped. The resulting ε-perturbed images were perceptually similar to the base image (Fig. 2e, left) but yielded large response changes (Fig. 2e, right, example neuron), consistent with the predictions of the compact model. We found this to be a strong effect across V4 neurons (Fig. 2f). Thus, despite practically infinite ways to slightly perturb an input image, the compact models correctly predicted two perturbations (ε-perturbed$^+$ and ε-perturbed$^-$) that strongly drove V4 responses in opposite ways. Taking the results of these experimental validations together (we also performed experimental validation on saliency maps; see Extended Data Fig. 5), we conclude that the compact models accurately capture the stimulus preferences of their V4 neurons, even for stimuli that lay well outside the distribution of training images of the model.

A consolidation step for diverse tuning

The high level of accuracy of the compact models in predicting held-out V4 responses and preferred stimuli motivated us to investigate their inner workings to gain more insight into the computations carried out by real V4 neurons. We started by comparing the number of filters for each layer as determined by our pruning algorithm. One possibility was that the number of filters would gradually increase from the earliest layer to the deepest layer, similar to the architectures of the most successful task-driven DNNs[35], as a higher dimensionality (that is, more filters) tends to make representations more linearly separable[36]. Another possibility was that the number of filters would vary greatly between compact models (where each model corresponds to one neuron), reflecting the varying complexity of models needed to explain the heterogeneity of stimulus preferences observed across different neurons (Fig. 2a). However, we found no evidence to support either scenario. Instead, we found that the early layers had the largest number of convolutional filters (Fig. 3a, layers 1–3), whereas the later layers had substantially fewer filters (Fig. 3a, layers 4–5 and spatial readout). This trend could not be explained by our method of pruning (Supplementary Fig. 3). We called the steep decrease in the number of filters between layers 3 and 4 (Fig. 3a; $P < 0.001$, paired permutation test, $n = 219$) as the 'consolidation step' because the model consolidates the information across many filters into only a few. That this abrupt consolidation occurred at the same layer for almost all models suggests that the consolidation step is a hallmark of V4 processing.

We wanted to explore why such a consolidation step would be needed. We reasoned that the early layers may represent a basis set of filters that provide a rich set of features useful for downstream neurons to specialize, much the way a bank of Gabor filters captures edge information useful for various vision tasks[37,38]. If true, we would expect to

see that the representations of the early layers are similar. To test this, we compared the internal activations of each layer between compact models using a similarity metric called the centred kernel alignment (CKA), commonly used to compare the inner representations of two deep neural networks[39]. A CKA similarity close to 1 indicates that the two layers have nearly identical representations, up to a linear rotation (see Methods). We found that the representations of the early layers were substantially more similar than those of later layers (Fig. 3b; the difference of medians between the similarities of layers 2 and 3 was significant, $P < 0.001$, paired permutation test); comparisons between the kernel weights of convolutional filters across models followed the same trend (Supplementary Fig. 3). The CKA similarity of outputs between any two models was low (Fig. 3b, output; the average signal correlation squared between models was $\rho^2 = 0.11$); when controlling for overlap in spatial receptive fields, the CKA similarity was even lower (Fig. 3b, layer 5 versus output). This suggests that each V4 neuron specializes in a unique way, consistent with the diversity in stimulus preferences (Fig. 2a).

That the compact models shared similar representations in the early layers was surprising, as presumably our V4 neurons, recorded with electrodes at least 400 μm apart, integrated input from separate populations of V1 and V2 neurons to build their feature selectivity. The similarity in representations suggested that we could predict all V4 neurons together with a backbone of shared 'core' layers; we sought to determine the smallest number of filters needed in these core layers. To do this, we used distillation to train 'shared' compact models and varied the number of filters in the first three core layers (Fig. 3c). Using our deep ensemble as a teacher model, we found that as few as 5 filters per core layer (that is, 15 filters total in layers 1–3) achieved prediction performance on par with ResNet50 features (Fig. 3c, leftmost point of the green trace intersects the brown dashed line), and that approximately 35 filters achieved prediction performance within 5% of the ensemble model (Fig. 3c, green arrow). Thus, high levels of compression are achievable when predicting a population of V4 neurons, suggesting V4 neurons share and reuse computations in early visual processing despite having diverse stimulus preferences.

Success in compressing the model to such a small number of filters was not trivially explained by similar tuning among the V4 neurons that we studied. These neurons had little-to-no signal correlations (Fig. 3b, output, and Extended Data Fig. 1), and the shared compact models captured diverse feature preferences (Supplementary Fig. 3). In addition, we found similar amounts of large compression for V4 datasets in other studies[20,40] (Extended Data Fig. 6), suggesting a generalizable principle for V4. The nonlinearities in the shared compact model were important: a model with the same architecture but only linear activation functions performed poorly (Fig. 3c, linear only). Task-driven DNN models themselves were also compressible (Fig. 3c, brown arrow, and Extended Data Fig. 6), as well as a data-driven compact model trained directly on V4 responses with no distillation (Fig. 3c, purple arrow). That these DNN models of V4 neurons are compressible is not straightforward: DNN units from task-driven DNNs were not nearly as compressible (Extended Data Fig. 6). Overall, these results suggest that the diverse specialization of V4 neurons likely arises during the consolidation step, whose precise readouts vastly differ between V4 neurons. Thus, understanding the computations occurring in the consolidation step is key to understanding the response profile of a V4 neuron.

Compact models of the visual cortex

Given our finding that a population of V4 neurons can be explained by a compressed set of computations, we wondered to what extent computations of other cortical areas along the visual stream were compressible. We trained shared compact models for three response datasets from other studies[40,41], including macaque V1, V4 and IT (Fig. 3d–f). Models for all three areas were highly compressible, with V1 being the most compressible (Fig. 3d, brown arrow, 5 filters), followed by V4 (Fig. 3e, brown arrow, 10 filters) and then IT (Fig. 3f, brown arrow, 60 filters), consistent with IT neurons having complex feature selectivity[42]. These results generalized to other task-driven DNNs (Extended Data Fig. 6); indeed, on public benchmark BrainScore[15], these compact versions outperformed their task-driven DNN counterparts while having orders of magnitude fewer parameters (Supplementary Table 2). Overall, our results suggest that at current prediction levels, the vast majority of filters in large task-driven DNN models are either spurious or redundant for predicting neural responses along the visual stream. Our compact models balance prediction and parsimony; we next investigated whether this parsimony leads to human-digestible explanations of visual processing.

The computations of a dot detector in V4

The small size of a compact model motivated us to explain its inner workings. We adopted an approach inspired by how visual neuroscientists probe the tuning of V4 neurons to object boundaries, shapes, colour, orientation and texture[43]. We focused on the salient specialization of dot-selective V4 neurons, observed in our own recorded neurons (Fig. 2a) and in other studies[20,27,30,31]; we confirmed that a subset of V4 neurons had preferences for dots in closed-loop validation experiments (Extended Data Fig. 7). Although the presence of dot detectors in V4 is interesting in its own right (possibly representing a visual bias towards detecting facial features such as eyes[31]), we were more interested in the computations that occur between the retinal inputs and the responses of a V4 dot detector. Thus, we chose to examine a compact model whose preferred stimuli greatly emphasized dots (Fig. 4a) and varied in response to dot size (Fig. 4b); this model was also selective for dot location and number (Extended Data Fig. 8). We confirmed that dot detection was a main computation of this model, as a training diet comprising only artificial dot stimuli still resulted in good prediction of the responses of the corresponding V4 neuron to natural images (Extended Data Fig. 8).

One might expect that a dot detector comprises convolutional filters with centre-surround receptive fields to recognize differently sized dots, but this was not the case for our example dot detector (Extended Data Fig. 8). Thus, we wanted to explore where in the model dot size selectivity arises. We ablated each filter separately and measured whether the output of the model became invariant to dot size by devising a metric called the dot size invariance (DSI) index. A DSI close to 1 indicates that a filter is necessary for dot size selectivity (Fig. 4c, filter L4F1 ablated), whereas a DSI close to 0 indicates that a filter, ablated or intact, contributes little to dot size selectivity (Fig. 4c, filter L3F33 ablated). We found that despite each single filter in layer 3 having little-to-no contribution to dot size selectivity (Fig. 4d, layer 3, DSI < 0.25), a filter in layer 4 strongly contributed to dot size selectivity (Fig. 4d, L4F1, DSI ≈ 1.0). This suggests that dot detection emerges after the consolidation step by reading out activity from multiple layer 3 filters. We isolated these filters involved in the readout by cumulatively ablating filters in layer 3 while DSI remained small; eventually DSI must increase, revealing the contributing filters (see Methods). Only 10 filters in layer 3 were strong contributors to dot size selectivity (Fig. 4e, rightmost dots); the other weak-contributing filters (Fig. 4e, leftmost dots) served to establish selectivity for other features, including the number of dots (Extended Data Fig. 8).

We further investigated the 6 most salient of the 10 contributing filters in layer 3 by observing their responses to an input image of a small dot. Four of these filters received input activity detecting curvature at four locations around the edge of the dot (Fig. 4f, layer 3 input column, top four filters); their excitatory kernel weights led to output activity with large positive values (Fig. 4f, layer 3 convolved output column, top four rows). The remaining two filters were inhibitory for large, oriented edges in their input (Fig. 4f, layer 3 filters column, bottom two rows) and responded weakly to a small dot devoid of large

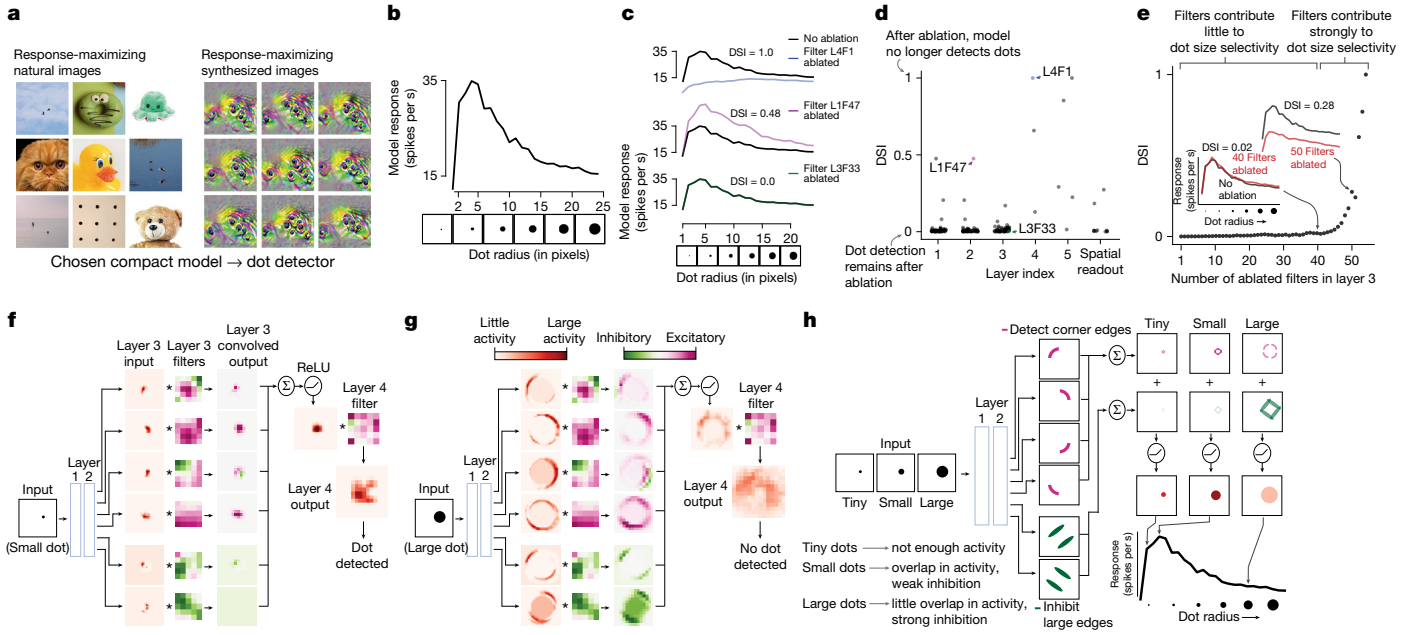


Fig. 4 | Uncovering the computations of a dot-detecting compact model. **a**, Response-maximizing images for a chosen 'dot-detecting' compact model. The left nine images were reproduced from Adobe Stock. **b**, Responses of this dot-detecting model to artificial dot stimuli varying in dot size. The same model was selective for dot location and number (Extended Data Fig. 8). **c**, Model responses for either no ablation (black lines) or ablated filters (coloured lines) to the same artificial dot stimuli as in panel **b**. An ablated filter ('L4F1' stands for layer 4, filter 1) has its kernel weights set to 0. We assessed the effect of ablation via DSI that measures the extent to which the model varies its response to dot size after filter ablation (see Methods). **d**, DSI for each individual convolutional filter of the chosen model. The coloured dots correspond to example filters in panel **c**. **e**, Cumulatively ablating filters in layer 3 to identify the subset of filters that contribute to dot size selectivity. We chose the next filter that, once ablated, led to the smallest increase in DSI. This filter remained ablated, and in a greedy manner, we proceeded to choose the next filter to ablate. After enough ablations, DSI must increase, revealing the contributing filters (rightmost dots). **f**, Investigating the activity of layer 3 filters that strongly contribute to dot size selectivity (that is, six among the rightmost dots in panel **e**) for input with a small dot. The input activity map for each layer 3 filter (layer 3 input) represents the processed image after the first two layers; each activity map is then convolved by its corresponding layer 3 filter (layer 3 filters) to produce the convolved output (layer 3 convolved output). This output is then summed element-wise across filters, passed through a ReLU, and then convolved again by a layer 4 filter to produce the processed output activity (layer 4 output). For input of a small dot, the layer 4 output activity is large (dark red), indicating that a dot was detected. **g**, Analysing the same filters as in panel **f** except for an input image with a large dot; the output activity (layer 4 output) is low (light red), indicating that no dot was detected. **h**, Illustrative diagram describing the computations in the dot-detecting compact model. The four excitatory filters (pink) detect the four corner edges of the dot, whereas the two inhibitory filters (green) detect large edges. A tiny dot leads to weak excitatory activity, whereas a large dot has little overlap in excitatory activity but strong inhibition. A small dot has strong overlap in activity and weak inhibition, leading to a large response.

edges. The strong overlap in positive activity across the top four filters and the weak inhibition by the bottom two filters overall yielded large positive activity (Fig. 4f, layer 4 output), which, after pooling across spatial locations, led to the result of a dot being detected. For an input image of a large dot, the story changed (Fig. 4g). The output activity of the top four filters no longer spatially overlapped (Fig. 4g, top four rows), and the bottom two filters provided strong inhibition due to the prominent edges of the large dot (Fig. 4g, bottom two rows). The end result was a low level of output activity (compare layer 4 output between Fig. 4f,g) and no dot being detected. We confirmed that other dot-detecting compact models operated with almost identical computations (Extended Data Fig. 9).

We condensed these results into a diagram of how this compact model operates as a dot detector (Fig. 4h); the important computations occurred in the readout of layer 3, consistent with our hypothesis that a V4 neuron specializes via its consolidation step. We also investigated the preference of this compact model for multiple dots, and found this selectivity largely emerges from filters in layers 4 and 5 whose large magnitude weights along their perimeters favour distances between dots (Extended Data Fig. 8).

Discussion

Despite efforts in model compression and pruning[17,25], the best-performing computer vision models at object recognition on ImageNet remain complex, comprising millions of parameters[13]. Similarly, scaling up the number of parameters has been a key factor in the success of large language models[14]. Although our work demonstrates that model compression is possible for predicting 200 visual cortical neurons together, this may not hold true when predicting orders of magnitude more neurons with diverse feature selectivity. In addition, the repeat-to-repeat variability present in neural responses may conceal subtle computations that would otherwise increase model complexity (although this appears not to be the case for our data; see Extended Data Fig. 6). It is thus possible that an understanding of the visual cortex in terms of conceptual, human-digestible diagrams (much like the Hubel–Wiesel model of V1) will not fully capture the complexity of visual processing in the brain. However, such diagram-like models are invaluable for providing readily available, specific and testable hypotheses about the underlying computations of the brain. For example, experimental observations and simple diagrams of divisive normalization[44,45] set the stage for sophisticated computational models[46,47] and experiments[48,49]. Our own analysis of a dot-detecting compact model suggests an underlying computation involving a balanced readout between excitatory corner filters and inhibitory large-edge filters (Fig. 4); this inspires new experiments with anatomical tracing[50], simultaneous recordings of V1/V2 and V4 neurons[51,52], and circuit perturbations[53,54] to test this hypothesis. Moreover, we propose model compression as a compromise between predictive power and conceptual understanding. Much the way a Taylor expansion approximates a function with increasingly

more complex polynomials ($f(x) \approx ax + bx^2 + cx^3 + \ldots$), our modelling framework provides a way to describe DNN models of the visual cortex with increasing levels of complexity. The fact that individual visual cortical neurons can be explained by such small compact models gives us hope that highly predictive, parsimonious models of the higher-order visual cortex are within reach.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-026-10150-1.

1. Yamins, D. L. & DiCarlo, J. J. Using goal-driven deep learning models to understand sensory cortex. *Nat. Neurosci.* **19**, 356–365 (2016).
2. Kriegeskorte, N. Deep neural networks: a new framework for modeling biological vision and brain information processing. *Annu. Rev. Vis. Sci.* **1**, 417–446 (2015).
3. Heeger, D. J. Half-squaring in responses of cat striate cells. *Vis. Neurosci.* **9**, 427–443 (1992).
4. Rust, N. C., Schwartz, O., Movshon, J. A. & Simoncelli, E. P. Spatiotemporal elements of macaque V1 receptive fields. *Neuron* **46**, 945–956 (2005).
5. Pillow, J. W. et al. Spatio-temporal correlations and visual signalling in a complete neuronal population. *Nature* **454**, 995–999 (2008).
6. Vintch, B., Movshon, J. A. & Simoncelli, E. P. A convolutional subunit model for neuronal responses in macaque V1. *J. Neurosci.* **35**, 14829–14841 (2015).
7. Ustyuzhaninov, I. et al. Digital twin reveals combinatorial code of non-linear computations in the mouse primary visual cortex. Preprint at *bioRxiv* https://doi.org/10.1101/2022.02.10.479884 (2022).
8. Maheswaranathan, N. et al. Interpreting the retinal neural code for natural scenes: from computations to neurons. *Neuron* **111**, 2742–2755 (2023).
9. David, S. V., Hayden, B. Y. & Gallant, J. L. Spectral receptive field properties explain shape selectivity in area V4. *J. Neurophysiol.* **96**, 3492–3505 (2006).
10. Yamins, D. L. et al. Performance-optimized hierarchical models predict neural responses in higher visual cortex. *Proc. Natl Acad. Sci. USA* **111**, 8619–8624 (2014).
11. Cadieu, C. et al. A model of V4 shape selectivity and invariance. *J. Neurophysiol.* **98**, 1733–1750 (2007).
12. Bahri, Y., Dyer, E., Kaplan, J., Lee, J. & Sharma, U. Explaining neural scaling laws. *Proc. Natl Acad. Sci. USA* **121**, e2311878121 (2024).
13. Liu, Z. et al. A ConvNet for the 2020s. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 11966–11976 (IEEE, 2022).
14. Kaplan, J. et al. Scaling laws for neural language models. Preprint at *arXiv* https://doi.org/10.48550/arXiv.2001.08361 (2020).
15. Schrimpf, M. et al. Integrative benchmarking to advance neurally mechanistic models of human intelligence. *Neuron* **108**, 413–423 (2020).
16. Hinton, G., Vinyals, O. & Dean, J. Distilling the knowledge in a neural network. Preprint at *arXiv* https://doi.org/10.48550/arXiv.1503.02531 (2015).
17. Han, S., Mao, H. & Dally, W. J. Deep compression: compressing deep neural networks with pruning, trained quantization and huffman coding. In *Proc. 4th International Conference on Learning Representations* (ICLR, 2016).
18. Butts, D. A. Data-driven approaches to understanding visual neuron activity. *Annu. Rev. Vis. Sci.* **5**, 451–477 (2019).
19. Pierzchlewicz, P. et al. Energy guided diffusion for generating neurally exciting images. In *Proc. Advances in Neural Information Processing Systems 36* (eds Oh, A. et al.) (NeurIPS, 2023).
20. Bashivan, P., Kar, K. & DiCarlo, J. J. Neural population control via deep image synthesis. *Science* **364**, eaav9436 (2019).
21. DiMattina, C. & Zhang, K. Adaptive stimulus optimization for sensory systems neuroscience. *Front. Neural Circuits* **7**, 101 (2013).
22. Pillow, J. W. & Park, M. in *Closed Loop Neuroscience* (ed. El Hady, A.) Ch. 1 (Elsevier, 2016).
23. Seung, H. S., Opper, M. & Sompolinsky, H. Query by committee. In *Proc. Fifth Annual Workshop on Computational Learning Theory* 287–294 (Association for Computing Machinery, 1992).
24. Cowley, B. & Pillow, J. W. High-contrast "gaudy" images improve the training of deep neural network models of visual cortex. In. *Proc. Advances in Neural Information Processing Systems 33* (eds Larochelle, H. et al.) (NeurIPS, 2020).
25. Frankle, J. & Carbin, M. The lottery ticket hypothesis: finding sparse, trainable neural networks. In *Proc. 7th International Conference on Learning Representations* 8954–8995 (ICLR, 2019).
26. He, Y., Zhang, X. & Sun, J. Channel pruning for accelerating very deep neural networks. In *Proc. IEEE International Conference on Computer Vision* 1398–1406 (IEEE, 2017).
27. Wang, T. et al. Large-scale calcium imaging reveals a systematic V4 map for encoding natural scenes. *Nat. Commun.* **15**, 6401 (2024).
28. Du, F., Núñez-Ochoa, M. A., Pachitariu, M. & Stringer, C. Towards a simplified model of primary visual cortex. Preprint at *bioRxiv* https://doi.org/10.1101/2024.06.30.601394 (2024).
29. Kamali, F., Suratgar, A. A., Menhaj, M. & Abbasi-Asl, R. Compression-enabled interpretability of voxelwise encoding models. *PLoS Comput. Biol.* **21**, e1012822 (2025).
30. Cowley, B. R., Williamson, R. C., Acar, K., Smith, M. A. & Yu, B. M. Adaptive stimulus selection for optimizing neural population responses. In *Proc. Advances in Neural Information. Processing Systems 30* (eds Guyon, I. et al.) (NeurIPS, 2017).
31. Willeke, K. F. et al. Deep learning-driven characterization of single cell tuning in primate visual area V4 unveils topological organization. Preprint at *bioRxiv* https://doi.org/10.1101/2023.05.12.540591 (2023).
32. Szegedy, C. et al. Intriguing properties of neural networks. In *Proc. 2nd International Conference on Learning Representations* (ICLR, 2014).
33. Guo, C. et al. Adversarially trained neural representations may already be as robust as corresponding biological neural representations. In *Proc. 39th International Conference on Machine Learning* (eds Chaudhuri, K. et al.) 8072–8081 (PMLR, 2022).
34. Berardino, A., Ballé, J., Laparra, V. & Simoncelli, E. Eigen-distortions of hierarchical representations. In *Proc. Advances in neural information processing systems 30* (eds Guyon, I. et al.) (NeurIPS, 2017).
35. Krizhevsky, A., Sutskever, I. & Hinton, G. E. ImageNet classification with deep convolutional neural networks. *Commun. ACM* https://doi.org/10.1145/3065386 (2017).
36. DiCarlo, J. J., Zoccolan, D. & Rust, N. C. How does the brain solve visual object recognition? *Neuron* **73**, 415–434 (2012).
37. Dapello, J. et al. Simulating a primary visual cortex at the front of cnns improves robustness to image perturbations. In *Proc. Advances in Neural Information Processing Systems 33* (eds Larochelle, H. et al.) (NeurIPS, 2020).
38. Federer, C., Xu, H., Fyshe, A. & Zylberberg, J. Improved object recognition using neural networks trained to mimic the brain's statistical properties. *Neural Netw.* **131**, 103–114 (2020).
39. Kornblith, S., Norouzi, M., Lee, H. & Hinton, G. Similarity of neural network representations revisited. In *Proc. 36th International Conference on Machine Learning* (eds Chaudhuri, K. & Salakhutdinov, R.) 3519–3529 (PMLR, 2019).
40. Majaj, N. J., Hong, H., Solomon, E. A. & DiCarlo, J. J. Simple learned weighted sums of inferior temporal neuronal firing rates accurately predict human core object recognition performance. *J. Neurosci.* **35**, 13402–13418 (2015).
41. Cadena, S. A. et al. Deep convolutional models improve predictions of macaque V1 responses to natural images. *PLoS Comput. Biol.* **15**, e1006897 (2019).
42. Conway, B. R. The organization and operation of inferior temporal cortex. *Annu. Rev. Vis. Sci.* **4**, 381–402 (2018).
43. Pasupathy, A., Popovkina, D. V. & Kim, T. Visual functions of primate area V4. *Annu. Rev. Vis. Sci.* **6**, 363–385 (2020).
44. Heeger, D. J. Normalization of cell responses in cat striate cortex. *Vis. Neurosci.* **9**, 181–197 (1992).
45. Carandini, M. & Heeger, D. J. Normalization as a canonical neural computation. *Nat. Rev. Neurosci.* **13**, 51–62 (2012).
46. Coen-Cagli, R., Kohn, A. & Schwartz, O. Flexible gating of contextual influences in natural vision. *Nat. Neurosci.* **18**, 1648 (2015).
47. Burg, M. F. et al. Learning divisive normalization in primary visual cortex. *PLoS Comput. Biol.* **17**, e1009028 (2021).
48. Ruff, D. A. & Cohen, M. R. A normalization model suggests that attention changes the weighting of inputs between visual areas. *Proc. Natl Acad. Sci. USA* **114**, E4085–E4094 (2017).
49. Verhoef, B.-E. & Maunsell, J. H. Attention-related changes in correlated neuronal activity arise from normalization mechanisms. *Nat. Neurosci.* **20**, 969–977 (2017).
50. Ungerleider, L. G., Galkin, T. W., Desimone, R. & Gattass, R. Cortical connections of area V4 in the macaque. *Cereb. Cortex* **18**, 477–499 (2008).
51. Semedo, J. D. et al. Feedforward and feedback interactions between visual cortical areas use different population activity patterns. *Nat. Commun.* **13**, 1099 (2022).
52. Jun, N. Y. et al. Coordinated multiplexing of information about separate objects in visual cortex. *eLife* **11**, e76452 (2022).
53. Day-Cooney, J., Cone, J. J. & Maunsell, J. H. Perceptual weighting of V1 spikes revealed by optogenetic white noise stimulation. *J. Neurosci.* **42**, 3122–3132 (2022).
54. Shahbazi, E., Ma, T., Pernuš, M., Scheirer, W. & Afraz, A. Perceptography unveils the causal contribution of inferior temporal cortex to visual perception. *Nat. Commun.* **15**, 3347 (2024).
55. Smith Breault, M. SciDraw: monkey brain. *Zenodo* https://doi.org/10.5281/zenodo.3926117 (2025).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026

Methods

Experimental animals

Three adult male rhesus macaques (*Macaca mulatta*: WE, PE and RA) were used for this study. Experimental procedures were approved by the Institutional Animal Care and Use Committee of Carnegie Mellon University and were performed in accordance with the United States National Research Council's Guide for the Care and Use of Laboratory Animals.

Neurophysiological experiments

Electrophysiology. We recorded extracellular activity from populations of V4 neurons in three awake, head-fixed monkeys; details about surgical procedures and electrophysiological recordings have been reported in a previous study[56]. In brief, animals were surgically implanted with a titanium headpost to allow for head fixation during experiments. We chronically implanted a 96-electrode array (Blackrock Microsystems; 1 mm in electrode length, 400 μm spacing in a 10 × 10 grid) in the left hemisphere V4 of each animal. V4 arrays were implanted on the prelunate gyrus medial to the inferior occipital sulcus. Electrodes of 1 mm in length would probably reach layers 4 and 5 of V4, where the cortical thickness is close to 2 mm; the precise layer of our recordings was not verified via histological analysis because after this study was completed the animals continued to be involved in additional data collection.

Processing neural responses. To process the recorded spike signals, we used an automated deep learning pipeline[57] built with custom Matlab software. This pipeline separates spike waveforms from noise on each channel; we took one neural 'unit' per electrode channel. We disregarded neural units with firing rates less than 1 spike per second. For each neural unit, we computed an unbiased estimator of the signal-to-noise ratio across images and repeats[58]; we disregarded units with the recommended signal-to-noise ratio of less than 0.15. Our data probably consisted of a mix of single and multi-units; we refer to each unit as a 'neuron' for simplicity. To confirm the reliability of the automated spike sorting, we also manually spike sorted the four held-out test sessions used to evaluate the models and found little difference in prediction performance between responses processed with automated versus manual spike sorting (Extended Data Fig. 2). The number of neurons across recording sessions and animals ranged from 21 to 89 neurons with a median of 58 neurons (Extended Data Fig. 1); exact numbers are in Supplementary Table 1. Similar to previous studies[10,20,59], we took spike counts in 100-ms time bins starting at 50 ms after stimulus onset to account for the visual response latency of V4 neurons.

Ideally, the same neurons would be present each session to give our model a large amount of training data to estimate the stimulus tuning of the neuron (a data-driven approach). However, although a subset of recorded neurons were probably the same across sessions for the same implanted array, we observed that after automatic spike processing, the number of neurons differed across sessions (Extended Data Fig. 1). This was expected due to a number of reasons, including slight shifts in the position of the array (caused by, for example, when the animal moved around in its home environment between sessions) or a build up of glial tissue over time around the implanted electrodes[60]. To retain the largest number of neurons possible for training, our modelling framework treated every neuron in each session as a potentially new neuron by giving each neuron its own linear mapping (Supplementary Fig. 1), the same approach used for task-driven models[1,15]. If the same neuron were present across multiple sessions, the linear mappings from DNN features to responses would be similar, effectively 'stitching' together the responses of a neuron across sessions[61].

Experimental task and data collection. To record responses to a large number of images, we trained each animal to perform a simple active fixation task. Each trial began with the presentation of a central blue fixation dot. The animal held an initial fixation for 150 ms. Following this, the animal maintained fixation while a sequence of six or eight images (depending on the animal) was presented. The square images had side lengths 11.2°, 10.6° and 8.0° and were centred at 12.7°/5.0°, 4.4°/6.2° and 3.8°/3.8° below/to the right of the central fixation dot for monkeys WE, PE and RA in order to overlap with the majority of the recorded receptive fields of V4 neurons. The average receptive field diameters for individual neurons were estimated to be 6.2°, 3.3° and 1.8° for monkeys WE, PE and RA (Supplementary Fig. 2). Each image was displayed for 100 ms with a 100-ms grey screen between images. Following this image sequence, the central fixation dot vanished and a target dot appeared approximately 10° from the location of the central fixation. To receive a liquid reward (that is, a 'correct' trial), the animal had to make a saccade to the target. Breaking fixation or failing to make a saccade to the target dot resulted in no reward and a 1-s time out before the next trial began. The gaze of the animal was tracked using an infrared eye tracking system (EyeLink 1000; SR Research) and monitored online by experimental control software to ensure fixation and to report the choice target of the animal. The display was a gamma-corrected flat-screen cathode ray tube monitor positioned 57 cm from the eyes of the animal with a resolution of 1,024 × 768 pixels, refreshed at a frame rate of 100 Hz. The background of the display was 50% luminance (grey). Stimulus generation was performed with PsychToolbox[62]. All animals performed the task well with many thousands of image flashes in each session.

Before each session, the set of images to show were chosen either randomly from a large dataset, by our models in a closed-loop manner, or by synthesizing images by optimizing model outputs (see 'Visual stimuli' section). The total number of unique images per session, ranging from 400 to 3,000 images and typically 2,000 images, was chosen to ensure enough repeats per image. All images were shown once in randomized order; for subsequent repeats, we showed the sequence of all images again but with a different randomized order. Each session lasted for multiple hours and ended when the animal showed signs of satiation. The number of repeats per image for each session ranged from 2 to 24 repeats; exact values are listed in Supplementary Table 1. To increase the number of repeats, we included any repeat for which the animal remained fixating on the central dot while the image was displayed; this included trials in which the animal may have broken fixation during a subsequent image, ending the trial and receiving no reward (an error trial). We discarded responses to images with only one repeat. The total number of sessions was 50 (7, 29 and 14 sessions for monkeys WE, PE and RA, respectively). Of these sessions, 44 sessions were used for model training; 1 session of randomly chosen natural images was used to validate model hyperparameters; 1 session was used to compare with responses from a previous study (Supplementary Fig. 2); and 4 sessions of randomly chosen natural images were used to test model predictions. These four test sessions consisted of one each from monkeys WE and RA and two from monkey PE (separated by 5 months). Images were unique across the validation and test sessions, and no model had access to any of these held-out images during training.

We note that although the electrode array may shift across sessions (and thus record from different neurons across sessions), a subset of the neurons in the test sessions were probably the same neurons in the training sessions. This was ideal for our purpose to build highly predictive data-driven models for our recorded V4 neurons (Fig. 1a) and why our experimental validation experiments were successful (Fig. 2). However, because we only recorded from approximately 200 V4 neurons across animals, we do not expect our deep ensemble to generalize well to V4 neurons on which the model has not been trained; we found the deep ensemble does no worse than its backbone task-driven DNN in predicting other neurons (Extended Data Fig. 2). We verified that our reported prediction performances, evaluated on all four test sessions (Fig. 1b), changed little when we removed either one of the sessions from the same animal. One potential concern is that neurons

recorded simultaneously in the same session may all have similar feature selectivity, leading to good prediction from a model that only needs to predict this redundant tuning. However, this is unlikely the case, as we found that neurons recorded within the same session had highly diverse feature selectivity (Fig. 2a) and small signal correlations (Fig. 3b and Extended Data Fig. 1) on par with or lower than previous studies (Supplementary Fig. 2). In addition, our prediction results were probably unaffected from noise correlations across simultaneously recorded neurons[5,63], as we only predicted repeat-averaged responses to images shown in random order across the session.

Visual stimuli

Our experiments involved many different types of images. These types included natural images from a large image database, images chosen by active learning, gaudy images[24], probe images used for experimental validation (Fig. 2), response-maximizing images generated from the deep ensemble model and the compact models, and artificial images (sinusoidal gratings and dot images, among others). All images were 112 × 112 pixels with RGB colour channels; this departs from previous work using greyscale images[20,59,64], but we found both types of images led to similar prediction performances (Supplementary Fig. 2). The types of images (and number of each type) shown for each session can be found in Supplementary Table 1—the total number of images per image type varied. We briefly describe each type of image here; examples of images are shown in Supplementary Fig. 4.

(1) Natural images: we defined natural images as those coming from our dataset of 12 million 'natural' colourful images subsampled from the Yahoo Flickr Creative Commons 100 Million Dataset (YFCC100M)[65], which contains close to 100 million images uploaded by users to Flickr between 2004 and 2014. Images are unlabelled (that is, no content information) and need not contain an object. After sampling an RGB image from YFCC100M, we first randomly cropped the image to have an equal number of row and column pixels and then downsized the image to 112 × 112 pixels. We chose YFCC100M primarily to ensure that our images used for training and testing were different from those in ImageNet[66], as the task-driven DNNs that we chose for features were trained on ImageNet images.

(2) Images chosen adaptively by active learning: in some sessions, a portion of the images were chosen adaptively via a closed-loop method called active learning (also known as optimal experimental design)[21,22,24,67–69]. To choose these images, we trained our deep ensemble model (see the 'Deep ensemble model' section) on all previous recording sessions. Then, for each of 500,000 candidate images randomly chosen from our image dataset, we computed the disagreement across ensemble members as the variance of predicted responses, averaged over neurons from the last recording session. We chose the images with the largest ensemble disagreement to show in the next session.

(3) Gaudy images: in our previous work, we identified a class of images that were adept at efficiently training DNN models of the visual cortex, even better (in simulations) than images chosen with active learning[24]. We called these images 'gaudy' images for their high-contrast edges and over-the-top colours. To generate a gaudy image, we first randomly chose an image from our image dataset. Then, for each pixel intensity, if that pixel intensity was greater than the average intensity across pixels for the same RGB channel, we set its value to 255; else 0. Thus, each pixel was one of 2^3 different colours (black, white, red, green, blue, and so on).

(4) Probe images used for validation experiments: to experimentally validate our compact model predictions, we probed each compact model in three different ways to identify its preferred stimuli. First, we identified the image selected from 500,000 candidate natural images that maximized the output of the compact model (Fig. 2b). Second, we synthesized the image via gradient methods that maximized the output of the compact model (Fig. 2c; see the 'Response-maximizing synthesized images' section). Third, we synthesized ε-perturbed images that slightly perturbed a natural 'base' image to either maximize or minimize the output response of the compact model (Fig. 2d–f; see the 'Experimental validation of the compact models' section).

(5) Response-maximizing natural and synthesized images of the deep ensemble model: we identified response-maximizing natural images and response-maximizing synthesized images for the deep ensemble model in the same way that we identified the probe images for the compact models.

(6) Artificial images: in a small number of sessions, we presented artificial stimuli. For one session, we presented static sinusoidal gratings varying in orientation (from 0° to 162°, 14 equally spaced orientations in total) and spatial frequency (from 0.02 pixels per degree to 0.5 pixels per degree, 14 equally spaced frequencies in total). In another session, we presented 'dot' images, which varied in dot location, dot size and number of dots (Extended Data Fig. 7).

Predicting V4 responses

Given a set of model features taken either from a task-driven or data-driven DNN or our deep ensemble model (described in later sections), we estimated the extent to which the features predicted V4 responses via a linear mapping. We followed recent work that uses a factorized linear mapping[20,70], taking advantage of decomposable spatial and channel information. Specifically, consider the output features (or activity maps) of a convolutional layer for one image as a tensor X of shape $p \times p \times K$ for p pixels and K channels/convolutional filters. Fitting the weights of a linear regression to one V4 neuron would result in weight tensor W of shape $p \times p \times K$. With a factorized linear mapping, we first spatially read out each activity map with the same spatial filter (W_{spatial} with shape $p \times p$) and then linearly combined these readouts across filters (W_{mixing} with shape $K \times 1$) via the following equation:

$$\hat{y}(X) = \sum_k W_{\text{mixing}}[k] \sum_{i,j} W_{\text{spatial}}[i,j] X[i,j,k]$$

This factorized linear mapping reduces the number of weights to $p \times p + K$; in other words, W_{spatial} and W_{mixing} together form a low-rank approximation of W, the weight tensor from standard linear regression. We regularized the weights of the factorized linear mapping with an L2 penalty and fit the weights using stochastic gradient descent (Adam optimizer) with early stopping. Values for the hyperparameters (α for the L2 penalty, learning rate, among others) were identified with data from the validation session. We confirmed that our fitting procedure of the factorized linear mapping outperformed ridge regression (Extended Data Fig. 1) and reproduces previously reported prediction performance on publicly released datasets of V4 responses[20,59] (see Extended Data Fig. 2).

For our test data, we held out 4 recording sessions for which we presented 1,200–1,600 colourful, natural images per session (images were unique across sessions) with an average of 10 repeats per image (see Supplementary Table 1). For each session, we used eightfold nested cross-validation; we trained the factorized linear mapping on three-quarters of the held-out image–response pairs, used one-eighth of the remaining image–response pairs as validation data for early stopping, and computed the predicted responses for the last remaining 1/8 images (test data). We then concatenated the predicted responses of the test data for all cross-validation folds as one vector $\hat{\mathbf{r}} \in \mathcal{R}^M$ for all M images of the session. This procedure is standard practice for robust performance measurements; however, simply splitting the data into a training, validation and test set (onefold cross-validation) yielded similar performance results even for 100 test images (Supplementary Fig. 2). This suggests that the number of neurons, number of images and noise levels of our data are appropriate for robust performance measurements.

To assess prediction performance, standard practice is to account for repeat-to-repeat variability by normalizing the explained variance between the model and repeat-averaged responses $R^2_{\text{model vs data}}$ by the explained variance between two sets of repeat-averaged responses where repeats are split into two groups $R^2_{\text{split1 vs split2}}$[10,15]. The $R^2_{\text{split1 vs split2}}$ can be thought of as a noise ceiling for the largest possible value of $R^2_{\text{model vs data}}$. Recent work[58] has proposed noise-corrected R^2_{ER} as a statistically consistent R^2 estimator of the expected response (ER) that is more robust than other R^2 metrics, especially for small numbers of images and repeats. We compared the values of R^2_{ER} and those of BrainScore R^2 (another estimator of noise-corrected R^2)[15] for our V4 responses, and found little difference between the two (Extended Data Fig. 2), suggesting that the number of test images (greater than 1,000 images per test session) and repeats (an average of 10 repeats per image) were large enough that the choice of estimator had small effects on our R^2 estimates.

In this work, we used R^2_{ER} as our measure for noise-corrected R^2. We computed R^2_{ER} with the following procedure. For the repeat-averaged response $\mathbf{r}_i$ of a neuron and the predicted response $\hat{\mathbf{r}}_i$ of the model to the i-th image, the R^2_{ER} is computed as follows:

$$\mathbf{z}_i = \mathbf{r}_i - \frac{1}{M}\sum_{j=1}^{M}\mathbf{r}_j, \qquad \hat{\mathbf{z}}_i = \hat{\mathbf{r}}_i - \frac{1}{M}\sum_{j=1}^{M}\hat{\mathbf{r}}_j$$

$$\text{noise} - \text{corrected } R^2_{\text{ER}} = \frac{\left(\sum_{i=1}^{M}\hat{\mathbf{z}}_i \cdot \mathbf{z}_i\right)^2 - \sigma^2/K \cdot \sum_{i=1}^{M}\hat{\mathbf{z}}_i^2}{\sum_{i=1}^{M}\hat{\mathbf{z}}_i^2 \cdot \sum_{i=1}^{M}\mathbf{z}_i^2 - \sigma^2/K \cdot (M-1) \cdot \sum_{i=1}^{M}\hat{\mathbf{z}}_i^2}$$

where M is the number of images and K is the number of repeats per image. The left terms in both the numerator and denominator are the same as found in computing standard R^2, and the right terms in both the numerator and the denominator subtract off bias arising from repeat-to-repeat variability σ^2. Because different images may have a different number of repeats (depending on how many trials were completed by the animal), we took the unbiased estimate of σ^2 for each image and averaged across images: $\hat{\sigma}^2 = \frac{1}{M}\sum_{i=1}^{M}\frac{1}{K_i - 1}\sum_{j=1}^{K_i}(\mathbf{r}_{ij} - \mathbf{r}_i)^2$, where K_i is the number of repeats for the i-th image and $\mathbf{r}_{ij}$ is the response for the i-th image and j-th repeat. We set K to the average number of repeats across images. We refer to R^2_{ER} as noise-corrected R^2 in the rest of the paper.

Task-driven DNN models

We predicted held-out V4 responses with task-driven DNNs (Fig. 1b). Previous studies have found that the internal layers of task-driven DNNs trained to perform object recognition on ImageNet[66] to be predictive of V4 responses[1,20,71,72]. We chose a set of task-driven DNNs readily available to download in Keras[73] as well as some that have shown high prediction on public benchmarks[15], available in PyTorch[74]; all modelling and analysis code was in Python. Each DNN differed in architecture, optimization and classification performance on ImageNet. Input images were resized and processed according to the data pre-processing procedure of each DNN. To identify the layer used to predict V4 responses for each DNN, we computed the cross-validated noise-corrected R^2 between the activity of each layer and V4 responses from the validation recording session using ridge regression and chose the layer with the largest R^2 (see the 'Predicting V4 responses' section). The chosen layer was typically one of the middle layers, consistent with previous studies[10,15,59]. The task-driven DNNs and identified layers (using the names as defined in Keras) were as follows: ResNet50, activation_33 (ref. 75); MobileNet, conv_pw_9_relu[76]; DenseNet169, pool3_pool[77]; Xception, add_7 (ref. 78); InceptionResNetV2, mixed_6a[79]; VGG19, block4_pool[80]; and NASNetLarge, activation_128 (ref. 81). We considered three additional DNN models shown to be predictive of V4 responses, accessible in PyTorch: AlexNet, features[7][35]; CORnet-S, module V4 output[82]; and ResNet50-robust ($L_2\,\epsilon = 0.1$), layer3[0][83]. We have reported results for the task-driven DNNs with the top 10 prediction performances, but we also evaluated other task-driven DNNs (Extended Data Fig. 2). We found that the average noise-corrected $R^2 \approx 0.45$ of these task-driven DNN models (Fig. 1b and Extended Data Fig. 2) was comparable with that reported in previous studies[10,15,84]. An untrained task-driven model (ResNet50 with randomly initialized weights) had a much worse noise-corrected $R^2 = 0.13$, suggesting that training on a relevant task (in this case, object recognition) is needed for good prediction of higher-order cortical neurons in the primate visual cortex.

Data-driven DNN models

One advantage of our V4 response dataset is the large number of recording sessions, comprising responses to tens of thousands of images, that may be used for training data-driven DNN models of V4 neurons. We wondered whether the amount of recording sessions was enough to train randomly initialized data-driven DNNs to achieve prediction levels on par with task-driven DNNs[19]. To this end, we trained a data-driven DNN with 5 separable convolutional layers (200 filters per layer for the first three layers and 100 filters per layer for the last two layers) on the 44 training recording sessions (Fig. 1b, 'single DNN'); the architecture and training procedure were identical to those of the data-driven shared compact models (see the 'Compact models shared across neurons' section and the architecture of the distilled model; Supplementary Fig. 1). We also considered an ensemble of 25 data-driven DNNs (Fig. 1b, data-driven 'ensemble'); each ensemble DNN had the same architecture and training data as the single DNN above, but each had a different random initialization. The output prediction was the mean predicted response averaged across ensemble members.

Hybrid deep ensemble model

Our deep ensemble model (Fig. 1a) was a hybrid model, combining a task-driven DNN (via transfer learning) and a data-driven DNN (an ensemble of DNNs). To compute the predicted response for an image, the image was first processed and fed through ResNet50 until layer activation_33, which outputs a $14 \times 14 \times 1{,}024$ tensor of activity maps (each with shape 14×14) for 1,024 channels. This feature tensor was fed as input into an ensemble of small DNNs; each ensemble DNN received the same feature tensor as input. Each ensemble DNN, trained separately, outputs a vector of predicted responses for n neurons; these vectors were averaged across ensemble members to get the final predicted response vector.

Architecture of ensemble DNN. We briefly describe the architecture of the ensemble DNN; Supplementary Fig. 1 contains a detailed diagram. The input is a $14 \times 14 \times 1{,}024$ feature tensor from a middle layer of ResNet50. The first layer of the ensemble DNN reduces the number of input channels from 1,024 to 512 by taking linear combinations of the input channels (that is, a kernel shape of 1×1); this is followed by batchnorm[85] and ReLU activation operations. The second layer is a separable 2D convolution[76] of 512 filters with kernel shape 3×3 and a stride of 2. The output of layer 2 is a $7 \times 7 \times 512$ tensor. Layers 3–6 have the same following skip connections, inspired by ResNet[75]. This comprises (1) a separable 2D convolution (256 filters with kernel shape 3×3) followed by batchnorm and ReLUs, (2) a separable 2D convolution (256 filters with kernel shape 3×3) followed by batchnorm and ReLUs, and (3) a convolutional layer to expand the dimensionality (512 filters with kernel shape 1×1). The output, a tensor of shape $7 \times 7 \times 512$, was added to the input of the layer as a skip connection, and the summed tensor was fed into the next layer. The final output embedding (the output of layer 6) is a tensor with shape $7 \times 7 \times 512$. This embedding was passed through a factorized linear mapping that spatially pooled and then linearly combined the activity maps to get predicted V4 responses (see 'Predicting V4 responses'). We chose this architecture and hyperparameters after a large architecture search with the

validation recording session; we found that the number of layers, nonlinearities and number of ensemble DNNs were important contributors to prediction performance (Extended Data Fig. 1).

Training optimization details. We trained the deep ensemble model on 44 recording sessions. Each recording session had its own factorized linear mapping trained separately from any other session, following previous work using a backbone DNN with neuron-specific readouts[86,87]; all other model weights were shared across sessions (see Supplementary Fig. 1). Responses for each V4 neuron were repeat averaged and then z-scored to balance gradients when predicting multiple V4 neurons. We optimized the model using stochastic gradient descent with learning rate $\eta = 2$, momentum $\nu = 0.7$ and a batch size of 64; optimizing with Adam[88] did not perform well when training across sessions. After a pass over all 44 recording sessions (an epoch), we decayed the learning rate by a factor of $\delta = 0.85$ and incremented the number of times training was performed for each session (for example, for the third epoch, we would load the data of a session into memory and perform three training passes on that session before moving to the next training session). Before training on any single session, the factorized linear mapping weights for that session were randomly initialized with values close to 0 (instead of saving weight values from previous epochs), which encouraged large gradients to update the shared weights. The final model was chosen via early stopping by taking the model with the best prediction performance (evaluated on a validation recording session) across training epochs.

Closed-loop model training. Our deep ensemble model was highly predictive of V4 responses to held-out images with a median noise-corrected $R^2 = 0.62$ (median $R^2 = 0.59$, 0.64, 0.62 and 0.62 for the four held-out recording sessions from monkeys WE, PE, PE-2nd experiment and RA, respectively). We wondered to what extent this high prediction accuracy stemmed from training the deep ensemble model in closed-loop experiments in which the model chose the next set of stimuli to show that best refined its parameters (called active learning or optimal experimental design). We trained the ensemble model (five ensemble DNNs) on images adaptively chosen by the deep ensemble model in the closed-loop experiments as well as gaudy images (see the 'Visual stimuli' section); as a control, we also trained the same model on randomly chosen natural images (Fig. 1c). We only considered training sessions in which both types of images were shown (30 sessions total). We equalized the number of images for both types by random sampling and trained the ensemble model on each type separately (using a learning rate $\eta = 1.5$ to account for the smaller amount of training data). We varied the number of training sessions by taking random subsets of the 30 sessions, training a new ensemble model for each subset. We tested prediction performance on the same four held-out sessions used for assessing task-driven and data-driven DNNs (Fig. 1b); these held-out sessions only had responses to randomly chosen natural images. In other words, training on active learning or gaudy images while testing on natural images represents an out-of-distribution prediction problem; only if these active learning or gaudy images more strongly emphasize important features of natural images would training on them achieve better prediction performance than that of training on randomly chosen natural images[24].

Identifying compact models

Ideally, we would start with a compact DNN model and directly fit its weights to the real data. However, when the amount of training data is limited (as in our case), this often leads to overfitting; indeed, training a small DNN directly on V4 responses underperformed versus task-driven DNNs (Fig. 1b). One way a large DNN model, trained on the same data, overcomes this overfitting is by having access to many subnetworks or 'lottery tickets'[25], which we also observed (Figs. 1b and 3c and Extended Data Fig. 6). Through pruning techniques[17,26,89–91], one can identify a small subnetwork (or winning ticket) that retains the same prediction performance as the large model. Motivated by this approach, we performed model compression on our highly predictive deep ensemble model to find a 'compact' model that achieves high prediction performance with as few parameters as possible.

We sought to identify one compact model for each held-out V4 neuron (219 models in total) to ensure all filters in a single model contributed solely to the corresponding neuron; we later relaxed this assumption by training compact models whose filters were shared across neurons (Fig. 3c). We first obtained the predicted responses from the deep ensemble model to 12 million images from our image dataset. For each held-out V4 neuron, we trained a factorized linear mapping (see the 'Predicting V4 responses' section) from the output embeddings of the deep ensemble model to V4 responses using half of the image–response pairs for that neuron's session; the other half was held out to measure prediction performance (noise-corrected R^2s in Fig. 1e). We then performed two model compression techniques used in deep learning: knowledge distillation and pruning. Knowledge distillation trains a single 'student' DNN to predict the averaged output of an ensemble of DNNs (the 'teacher' model)[16]; the single, small DNN overcomes overfitting because unlike the relatively small number of image–V4 response pairs (typically 2,000 per session), we trained the distilled DNN on ensemble responses to 12 million images. Pruning takes a trained DNN, ablates unnecessary weights (that is, set their values to 0) and retrains[17,26,89–91]; DNNs with up to 90% of their weights pruned have been found to be as predictive as the original unpruned DNN[25]. We used both of these techniques to identify compact models.

We first performed distillation on our ensemble model. We initialized a 'distilled' DNN model with the following architecture. The (recentred) input RGB image has shape $112 \times 112 \times 3$. The first layer is 2D convolutional comprising 100 filters with kernel size 5×5 followed by batchnorm and ReLUs. Layers 2–5 are each 2D separable convolutions comprising 100 filters with kernel size 5×5 followed by batchnorm and ReLUs; layers 2 and 3 take a stride of 2. The final dense 'spatial readout layer' linearly maps the output activity of layer 5 (of shape $28 \times 28 \times 100$) to the scalar V4 response. We chose all layers to have 100 filters after a hyperparameter sweep for the number of filters per layer. We trained the distilled model on one pass of 12 million image–response pairs, where responses were the predicted responses from our ensemble model, using Adam[88] to minimize the mean squared error with learning rate $\eta = 10^{-4}$. For every 500,000 training images, we smoothed the kernel weights of layers 2–5 and the spatial readout layer with a Gaussian filter ($\sigma = 0.5$ pixels) for better interpretability of the kernel weights.

The distilled models were substantially smaller than our deep ensemble model (600 versus 150,000 filters). However, we wondered whether different V4 neurons could be explained by even fewer filters than those of the distilled models. For example, an edge-detecting V4 neuron may need fewer filters than a curve-detecting V4 neuron. To test this, we developed a channel-wise pruning technique that ablated entire filters (as ablating individual weights could possibly lead to many filters with sparse weights, making the non-smooth kernel weights difficult to interpret). We developed and performed the following pruning technique, inspired by hard filter pruning from previous studies[90,91]. We first passed 5,000 randomly chosen natural images as input and collected the output activity for each convolutional layer and the spatial readout layer. Then, for each layer, we ordered the filters based on the variance of the activity of each filter summed over spatial locations. We then identified the largest subset of lowest-variance filters that, once ablated, still explains at least 90% of the total variance of the original output activity for that layer. In other words, we removed filters with either weak activity or whose activity is unlikely to reach downstream layers. We performed this procedure first for the spatial readout layer (that is, the deepest layer), followed by each preceding layer until layer 1 was reached; reversing this layer order (pruning early to late layers) led to pruned models with, on average, 100 extra filters (Supplementary Fig. 3).

We note that the number of convolutional filters in layers 1 and 2 (Fig. 3a) must be equal by definition as a consequence of having a convolutional layer followed by a separable convolutional layer (Supplementary Fig. 1). We then retrained this pruned network in the same way as the distilled network with one pass of training over the responses of the deep ensemble to 12 million images. We call the retrained pruned network a 'compact' model.

Response-maximizing synthesized images

To identify the visual features that a compact model prefers, one approach is to search for the image that maximizes the output response of the model. A simple search involves computing the output response of the model for a large pool of candidate images and then choosing the image that evokes the largest output[30]. However, because the pool of candidate images is finite, not all possible images are considered. A more expressive search is to synthesize the maximizing image via gradient ascent[20,64]. This takes advantage of the fact that the mapping of the compact model is entirely differentiable. The search begins with a white noise image where the value of each pixel intensity is drawn from a Gaussian with mean of 128 and standard deviation of 50 (clipped between 0 and 255). This initial image was passed through the model to compute its output. We then took the gradient of the output with respect to the input image and passed this gradient back through the network (akin to backpropagation), updating the image with the gradient ascent rule: $\mathbf{x}_{\text{next}} = \mathbf{x}_{\text{current}} + \eta \nabla \mathbf{f}(\mathbf{x})$ for image $\mathbf{x}$, gradient of model output $\nabla \mathbf{f}(\mathbf{x})$ and step size η (in our analyses, $\eta = 10$). We performed this for 1,000 steps or until the maximized output did not change. Because moving along the gradient direction may lead to a synthesized image far outside the natural image manifold, we smoothed the gradient at each step (Gaussian filter with $\sigma = 1$) and smoothed the image (Gaussian filter with $\sigma = 1$) every 50 epochs. Extended Data Fig. 3 shows the maximizing synthesized image for each compact model and also includes discussion comparing our synthesized images to those of other studies[20,31,64].

Experimental validation of the compact models

We experimentally validated testable predictions of the compact models in follow-up experiments. We ran these experiments in two monkeys (PE and RA). We trained the deep ensemble model on any previously recorded sessions from any animal. We then identified a compact model for each V4 neuron from one of the held-out test sessions (for which only natural images were shown), following the same process as described in the 'Identifying compact models' section. We obtained three different predictions from the compact models. First, we identified the top response-maximizing natural image from 500,000 natural images for each compact model (Fig. 2b; 1 image for monkey PE and 10 images for monkey RA per compact model). Second, we identified the top response-maximizing synthesized image following the same process as outlined in the 'Response-maximizing synthesized images' section (Fig. 2c; 1 base image for monkey PE and 10 base images for monkey RA per compact model; each image began with a different white noise image). Third, we identified ε-perturbed images for each compact model (Fig. 2d–f; 12 images for monkey PE and 10 images for monkey RA per compact model). Similar in spirit to adversarial images[32,33,92,93] and eigen-distorted images[34], we defined an ε-perturbed image as a small ε perturbation to a base natural image that either maximizes (ε-perturbed$^+$) or minimizes (ε-perturbed$^-$) the response of the model. We computed each ε-perturbed image via gradient descent: starting with a natural image, we continued to perturb this base image via gradient descent until either 50 gradient updates were completed or the average intensity difference (absolute valued) between pixels of the initial natural image and the currently-perturbed image passed a threshold pixel intensity of 10 (where pixel intensities range from 0 to 255). Our threshold of an average pixel intensity difference of $\varepsilon = 10$ is an order of magnitude larger than that for adversarial images of deep neural networks ($\varepsilon = 1.66$ reported in refs. 32,92), whereas less than that for adversarial images of human perception ($\varepsilon = 32$ reported in ref. 93), placing our ε-perturbed images between DNN-perceivable and human-perceivable adversarial noise. We note that our compact models are still susceptible to adversarial noise, as the predictions of response magnitudes of the model to maximizing synthesized images (optimized in a way similar to adversarial images) were substantially larger than real V4 responses to the same images (Extended Data Fig. 5). We did not seek to optimize adversarial images (that is, optimizing for the smallest ε-perturbed image such that a statistically significant change in response from the base image occurs[33], as we were most interested in whether the compact models could be used to slightly perturb an image (out of an infinite number of possible perturbations) to elicit large changes in V4 responses.

We presented these probe images in succeeding sessions. Specifically, we showed response-maximizing natural and synthesized images together in the same session per animal; we showed ε-perturbed images in a separate session per animal. For comparison, we also presented a number of randomly chosen natural images within each session (250 images for monkey PE and 660 images for monkey RA). After recording V4 responses, we assessed whether the probe images did elicit changes in responses compared with the natural images. Because there was no guarantee that the recorded neurons on succeeding sessions would be the same as those targeted by the compact models, we first had to match each V4 neuron with a compact model. To do this, for every V4 neuron, we computed the noise-corrected R^2 between the responses of that neuron and the predicted responses of each compact model on all presented images except the response-maximizing or ε-perturbed images. The compact model that matched to a V4 neuron was the one with the largest R^2; any neuron with its largest $R^2 < 0.3$ was discarded (26 of 70 and 21 of 55 neurons for monkeys PE and RA, respectively). We then computed the repeat-averaged responses of each remaining neuron to the probe images for the matched compact model (Fig. 2b–f). The response-maximizing natural and synthesized images were shown in the same recording sessions (Fig. 2b,c), yielding 78 neurons combined over two animals; the ε-perturbed images (Fig. 2d–f) were shown in recording sessions separate from those with response-maximizing images, yielding 75 neurons combined over two animals. We confirmed that the response-maximizing natural and synthesized images did not simply excite all V4 neurons but rather targeted specific V4 neurons (Extended Data Fig. 4). We also confirmed that the randomly chosen natural images that elicited responses as large as the response-maximizing natural and synthesized images had similar image statistics (Extended Data Fig. 4).

CKA similarity

To compare the internal representations between two compact models (Fig. 3b), we used CKA similarity, a commonly used metric to compare representations between two deep neural networks[39]. CKA similarity assesses to what extent two representations—in our case, the hidden unit activations from a given layer between two compact models—match in correlation between (rotated) linear dimensions and in eigenspectra. CKA similarity is computed as follows. Consider the response activity of two representations $\mathbf{X} \in \mathcal{R}^{k_X \times m}$ and $\mathbf{Y} \in \mathcal{R}^{k_Y \times m}$ for the same m images, where k_X and k_Y are the number of feature variables in each representation (k_X and k_Y need not be equal). We assumed that $\mathbf{X}$ and $\mathbf{Y}$ are centred; the mean of each row is 0. We then defined CKA similarity (using a linear kernel as in ref. 39) as follows:

$$K(\mathbf{X}, \mathbf{Y}) = \frac{1}{(m-1)^2} \mathrm{Tr}(\mathbf{X}^T \mathbf{X} \mathbf{Y}^T \mathbf{Y})$$

$$\mathrm{CKA}_{\text{similarity}}(\mathbf{X}, \mathbf{Y}) = K(\mathbf{X}, \mathbf{Y}) / \sqrt{K(\mathbf{X}, \mathbf{X}) K(\mathbf{Y}, \mathbf{Y})}$$

where $\mathrm{Tr}(\cdot)$ denotes the trace operation. A CKA similarity close to 1 indicates that two representations are highly similar; a value close to 0 indicates that the representations are different.

CKA similarity has several nice properties, including invariance to orthogonal linear transformations (for example, if the rows of $\mathbf{X}$ were a permutation of the rows of $\mathbf{Y}$) and invariance to isotropic scaling (for instance, if $\mathbf{X} = c\mathbf{Y}$ for some scalar c). As opposed to a similarity metric based on canonical correlation analysis[94], CKA similarity will be larger if the eigenspectra of the two representations are similar—in other words, CKA similarity captures whether two representations have the same dominant dimensions. In addition, when both representations have one variable each, CKA similarity is equal to the squared Pearson correlation (that is, the signal correlation squared between the outputs of two models). Other similarity metrics exist[94,95]; however, because almost all of these similarity metrics rely on the same principles of invariance to linear transformations and isotropic scaling, we do not suspect our conclusions to differ across similarity metrics. We computed the CKA similarity between all pairs of models for each of the five layers, as well as the output of the model, in response to $m = 10{,}000$ randomly chosen natural images (Fig. 3b).

Compact models shared across neurons

Given our finding that the early layers of the compact models had similar convolutional filters (Supplementary Fig. 3) and similar representations across V4 neurons (Fig. 3b), we reasoned that we could identify a 'shared' compact model that predicted all 219 held-out V4 neurons together. Specifically, we wondered how many filters k were needed in the first three 'core' layers of this shared compact model. The architecture of the shared compact model resembled that of a compact model before pruning (that is, the 'distilled' model; see Supplementary Fig. 1). Namely, layer 1 was a 2D convolutional layer comprising k filters with kernel size $5 \times 5 \times 3$ (for 5×5 image patches and three RGB channels) followed by batchnorm and ReLUs. Layers 2 and 3 were 2D separable convolutional layers comprising k filters with kernel size 5×5 and a stride of 2 followed by batchnorm and ReLUs. Layers 4 and 5 were the same as layers 2 and 3 except always with 100 filters. The output embedding for m images is a tensor with shape $m \times 28 \times 28 \times 100$. We mapped this embedding to the responses of n neurons via factorized linear mappings (see the 'Predicting V4 responses' section). This architecture allowed us to vary the number of filters in the first three 'core' layers while allowing specialization of feature selectivity in layers 4 and 5. Our deep ensemble model served as a teacher model for distillation: for each of the four test sessions, we used half of the images to fit a linear mapping from the embeddings of the deep ensemble to the V4 responses. We split the remaining half of images as a validation set (for hyperparameter estimation and early stopping evaluation) and a test set for reporting noise-corrected R^2: the validation and test sets each comprised 300–400 images, more than enough for evaluating R^2 (Supplementary Fig. 2). We trained the shared compact model via distillation on predicted responses of our deep ensemble model (teacher model) to 12 million images using the Adam optimizer[88] with a learning rate of $\eta = 10^{-4}$, a gradient clip value of 0.5 and early stopping. We varied the number of filters k in the core layers and trained a new shared compact model for each k (Fig. 3c). We also confirmed that the response-maximizing synthesized images of the smaller shared compact models ($k = 50$) matched those of larger compact models ($k = 200$), indicating that smaller shared compact models were able to reproduce the stimulus preferences of the V4 neurons. To assess the extent to which nonlinearities in the shared compact model were important for prediction, we substituted all ReLU activation functions to linear identity functions and trained on the same ensemble responses (Fig. 3c, 'shared compact model linear only'). We note that the noise-corrected R^2 value reported for the deep ensemble in this analysis slightly differs from that in Fig. 1b because here we evaluated prediction performance on a single test set (onefold validation), whereas in Fig. 1b, we reported the cross-validated prediction performance.

To measure the extent to which a task-driven DNN model of V4 neurons is compressible, we treated the top-performing task-driven DNN, ResNet50-robust, as a teacher model (fitting a linear mapping between DNN features and V4 responses) for distillation of the shared compact model (the student model), training it on the predicted responses of ResNet50-robust to 12 million images (Fig. 1b). Other task-driven DNNs were also highly compressible (Supplementary Fig. 3). As a baseline, we also wondered how predictive a data-driven, non-distilled shared compact model trained on real V4 responses would be (Fig. 3c, purple trace). We used a similar training procedure as that for the deep ensemble model with a learning rate of $\eta = 10^{-1}$, a learning rate decay of 0.85 taken after each epoch and 5 passes per session for each epoch (see 'Training' in the 'Deep ensemble model' section), and early stopping evaluated on validation data. Neurons for each recording session had their own linear mapping private to that recording session. The shared compact model with 200 filters per core layer is the same as the data-driven DNN (Fig. 1b, 'single DNN'). One potential concern is that neurons recorded simultaneously may have similar stimulus tuning, trivially leading to high model compressibility; however, we ruled out this possibility, as both the real V4 neurons and the model V4 neurons had low signal correlations within a session (Extended Data Fig. 1 and Supplementary Fig. 2).

We further wondered whether the observed levels of model compression on our V4 response dataset generalized to other V4 datasets as well as other visual areas. We re-performed the same distillation and compact model training procedures for four publicly available datasets from other studies: macaque V1 (Fig. 3d; 115 neurons, 1,450 images and 4 repeats per image)[41], macaque V4 (Fig. 3e; 88 neurons and 3,200 images, repeat averaged)[40], macaque IT (Fig. 3f; 168 neurons and 3,200 images, repeat averaged)[40], as well as another macaque V4 dataset (Extended Data Fig. 6; 131 neurons, 640 images and 38 repeats per image)[20]. Each dataset had a unique set of images shown to each neuron; we split the set of images into a training, validation and test set, as we did for our own V4 dataset. For all 12 million colourful natural images used for training, we greyscaled and applied a circular mask (removing the surround) to better match the images presented in all four datasets. In addition, after distillation training on predicted responses, we fine-tuned the compact models on the real V4 responses in the training set and performed early stopping, evaluating on a held-out validation set different from the test set. We have reported raw R^2 for the V4 and IT datasets (Fig. 3e,f), as responses in these datasets were already repeat averaged (that is, no repeat-to-repeat variability to estimate a noise-corrected R^2). We have reported prediction performance for shared compact models of ResNet50-robust (Fig. 3d–f); we observed similar compression for other task-driven DNNs on these datasets (Extended Data Fig. 6). As a baseline, we trained data-driven shared compact models on the training sets. As expected, their prediction performance was much lower than that of the task-driven DNN (Fig. 3d–f, purple trace below the brown trace), because the amount of training data was small (1,000–2,000 images); we found increasing the amount of training data, as in our V4 experiments, closes this prediction gap (Figs. 1b and 3c).

Dot detector characterization

We assessed the preferred stimuli for each compact model (that is, the response-maximizing normal and synthesized images) and chose one that most resembled a dot detector. To ensure that the chosen compact model could be objectively described as a dot detector, we devised an artificial stimulus set in which we varied the dot location and size (Fig. 4b and Extended Data Fig. 8). For dot location, we varied the location of a dot with a small size (5-pixel radius) for 28×28 equally spaced locations; the preferred dot location of the model is the dot location with the largest response. For dot size, we placed a dot at the preferred location of the model and varied its radius from 2 pixels to 22 pixels. We also considered selectivity to dot number (Extended Data Fig. 8). In addition, we presented the same artificial stimulus set (varying dot location, size and number) in a recording session and found real V4 neurons that largely resembled dot detectors (Extended Data Fig. 7).

Finally, we verified that an artificial stimulus set of dots, gratings and plaids largely explained the response of the model to natural images (Extended Data Fig. 8), which strongly supports labelling this chosen compact model as a dot detector.

To identify which filters contributed to the selectivity of the chosen model to dot size (Fig. 4c–e), we considered the output responses $\mathbf{r}(x)$ of the model to the dot size stimuli x versus the output $\mathbf{r}_{\neg L\ell Fj}(x)$ of the model when we ablated the j-th filter from layer ℓ (that is, set its kernel weights to 0). If a filter contributed to dot size selectivity, we would expect to see a large change in the output of the model. To quantify this, we defined a DSI index for the j-th filter of layer ℓ as follows:

$$\mathrm{DSI}(\ell, j) = \frac{\sum_x [\tilde{\mathbf{r}}_{\neg L\ell Fj}(x) - \tilde{\mathbf{r}}(x)]^2}{\sum_x \tilde{\mathbf{r}}(x)^2}$$

where $\tilde{\mathbf{r}}(x)$ denotes the residual model response after the mean model response across all dot size stimuli was subtracted. We found this recentring to be necessary, as ablating some filters produced an additive offset change in response but otherwise left the dot size selectivity of the model intact. We set DSI = 0 for any value less than 0 (which occurs when $\tilde{\mathbf{r}}_{\neg L\ell Fj}(x)$ has larger variance or a different sign than $\tilde{\mathbf{r}}(x)$). A filter with DSI close to 1 indicates that, once ablated, the model output is no longer selective for dot size; in other words, the filter contributes strongly to the dot size selectivity of the model. A filter with DSI close to 0 indicates that the model retains its dot size selectivity whether this filter is ablated.

To isolate which filters in layer 3 most contributed to dot size selectivity (Fig. 4e), we cumulatively ablated layer 3 filters by the following iterative procedure. First, we identified the individual filter that, once ablated, yielded the smallest increase in DSI (that is, the filter that least contributed to the dot size selectivity of the model). This chosen filter remained ablated for the rest of the procedure. Then, in a greedy manner, we repeated this process for the remaining unablated filters. At some point, DSI must increase, as ablating all filters leads to DSI = 1. Thus, the filters chosen last in this process (the filters where DSI starts to increase) contribute most to the dot size selectivity of the model. We investigated the 10 filters to be chosen last (Fig. 4e, rightmost dots), from which we found 6 filters that carried out a simple computation (Fig. 4f–h); the remaining 4 filters were redundant or had weaker effects. We confirmed similar computational motifs in two other compact models that resemble dot detectors (Extended Data Fig. 9). We note that the filters that weakly contribute to dot size selectivity (Fig. 4e, leftmost dots) probably serve other computational purposes, such as filtering background textures, inhibiting visual features other than dots, or contributing to other feature selectivity including the number of dots. Indeed, we further investigated the preference for multiple dots and identified computational motifs that depended on the sparse perimeter weights of filters in layers 4 and 5 (Extended Data Fig. 8).

Statistical analysis

Significance testing was performed with two-tailed permutation tests on difference of medians unless otherwise stated. If the actual statistic was greater or less than all 'permuted' statistics under the null hypothesis, the P value was set to 0.001, equal to $1/n$ where the number of runs was $n = 1{,}000$. We applied paired permutation tests when comparing model performance (for example, Fig. 1b), as the neurons from the held-out test recording sessions remained the same. That the neurons were recorded simultaneously as well as across sequential sessions minimally influenced our prediction and compression results (see the 'Experimental task and data collection' section). Bootstrapped confidence intervals of medians (or quantiles estimated for the validation experiments in Fig. 2) were computed by 500 runs of re-estimating the median by sampling from the data with replacement. The numbers of images, repeats, neurons and sessions are listed in Supplementary Table 1.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Raw data including spike timing for all recording sessions are available[97]. Processed responses for model training and evaluation as well as stimulus images are available on GitHub (https://github.com/cowleygroup/V4_compact_models). Source data are provided with this paper.

Code availability

All spike sorting was performed using custom Matlab software available on GitHub (https://github.com/smithlabvision/spikesort). Model weights and code are available on GitHub (https://github.com/cowley-group/V4_compact_models).

56. Stan, P. L. & Smith, M. A. Recent visual experience reshapes V4 neuronal activity and improves perceptual performance. *J. Neurosci.* **44**, e1764232024 (2024).
57. Issar, D., Williamson, R. C., Khanna, S. B. & Smith, M. A. A neural network for online spike classification that improves decoding accuracy. *J. Neurophysiol.* **123**, 1472–1485 (2020).
58. Pospisil, D. A. & Bair, W. The unbiased estimation of the fraction of variance explained by a model. *PLoS Comput. Biol.* **17**, e1009212 (2021).
59. Cadena, S. A. et al. Diverse task-driven modeling of macaque v4 reveals functional specialization towards semantic tasks. *PLoS Comput. Biol.* **20**, e1012056 (2024).
60. Sponheim, C. et al. Longevity and reliability of chronic unit recordings using the Utah, intracortical multi-electrode arrays. *J. Neural Eng.* **18**, 066044 (2021).
61. Degenhart, A. D. et al. Stabilization of a brain–computer interface via the alignment of low-dimensional spaces of neural activity. *Nat. Biomed. Eng.* **4**, 672–685 (2020).
62. Kleiner, M., Brainard, D. & Pelli, D. What's new in psychtoolbox-3? *Perception* **36**, 1–16 (2007).
63. Cohen, M. R. & Kohn, A. Measuring and interpreting neuronal correlations. *Nat. Neurosci.* **14**, 811–819 (2011).
64. Walker, E. Y. et al. Inception loops discover what excites neurons most using deep predictive models. *Nat. Neurosci.* **22**, 2060–2065 (2019).
65. Thomee, B. et al. YFCC100M: the new data in multimedia research. *Commun. ACM* **59**, 64–73 (2016).
66. Deng, J. et al. ImageNet: a large-scale hierarchical image database. In *Proc. 2009 IEEE Conference on Computer Vision and Pattern Recognition* 248–255 (IEEE, 2009).
67. Lewi, J., Butera, R. & Paninski, L. Sequential optimal design of neurophysiology experiments. *Neural Comput.* **21**, 619–687 (2009).
68. Benda, J., Gollisch, T., Machens, C. K. & Herz, A. V. From response to stimulus: adaptive sampling in sensory physiology. *Curr. Opin. Neurobiol.* **17**, 430–436 (2007).
69. DiMattina, C. & Zhang, K. Active data collection for efficient estimation and comparison of nonlinear neural models. *Neural Comput.* **23**, 2242–2288 (2011).
70. Klindt, D., Ecker, A. S., Euler, T. & Bethge, M. Neural system identification for large populations separating "what" and "where". In *Proc. Advances in Neural Information Processing Systems 30* (eds Guyon, I. et al.) (NeurIPS, 2017).
71. Richards, B. A. et al. A deep learning framework for neuroscience. *Nat. Neurosci.* **22**, 1761–1770 (2019).
72. Doerig, A. et al. The neuroconnectionist research programme. *Nat. Rev. Neurosci.* **24**, 431–450 (2023).
73. Chollet, F. et al. Keras. *Keras* https://keras.io (2015).
74. Paszke, A. et al. PyTorch: an imperative style, high-performance deep learning library. In *Proc. Advances in Neural Information Processing Systems* 32 (eds Wallach, H. et al.) (NeurIPS, 2019).
75. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. In *Proc. 2016 IEEE Conference on Computer Vision and Pattern Recognition* 770–778 (IEEE, 2016).
76. Howard, A. G. et al. Mobilenets: efficient convolutional neural networks for mobile vision applications. Preprint at *arXiv* https://doi.org/10.48550/arXiv.1704.04861 (2017).
77. Huang, G., Liu, Z., Van Der Maaten, L. & Weinberger, K. Q. Densely connected convolutional networks. In *Proc. 2017 IEEE Conference on Computer Vision and Pattern Recognition* 2261–2269 (IEEE, 2017).
78. Chollet, F. Xception: deep learning with depthwise separable convolutions. In *Proc. 2017 IEEE Conference on Computer Vision and Pattern Recognition* 1800–1817 (IEEE, 2017).
79. Szegedy, C., Ioffe, S., Vanhoucke, V. & Alemi, A. Inception-v4, inception-resnet and the impact of residual connections on learning. In *Proc. Thirty-First AAAI Conference on Artificial Intelligence* 4278–4284 (AAAI, 2017).
80. Simonyan, K. & Zisserman, A. Very deep convolutional networks for large-scale image recognition. Preprint at *arXiv* https://doi.org/10.48550/arXiv.1409.1556 (2014).
81. Zoph, B., Vasudevan, V., Shlens, J. & Le, Q. V. Learning transferable architectures for scalable image recognition. In *Proc. 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition* 8697–8710 (IEEE, 2018).
82. Kubilius, J. et al. CORnet: modeling the neural mechanisms of core object recognition. Preprint at *bioRxiv* https://doi.org/10.1101/408385 (2018).
83. Salman, H., Ilyas, A., Engstrom, L., Kapoor, A. & Madry, A. Do adversarially robust imagenet models transfer better? In *Proc. Advances in Neural Information Processing Systems 33* (eds Larochelle, H. et al.) (NeurIPS, 2020).

84. Zhuang, C. et al. Unsupervised neural network models of the ventral visual stream. *Proc. Natl Acad. Sci. USA* **118**, e2014196118 (2021).
85. Ioffe, S. & Szegedy, C. Batch normalization: accelerating deep network training by reducing internal covariate shift. In *Proc. 32nd International Conference on Machine Learning* (eds Bach, F. & Blei, D.) 448–456 (PMLR, 2015).
86. Lurz, K.-K. et al. Generalization in data-driven models of primary visual cortex. In *Proc. 9th International Conference on Learning Representations* (ICLR, 2021).
87. Turishcheva, P., Burg, M. F., Sinz, F. H. & Ecker, A. S. Reproducibility of predictive networks for mouse visual cortex. In *Proc. Advances in Neural Information Processing Systems 37* (eds Globerson, A. et al.) (NeurIPS, 2024).
88. Kingma, D. P. & Ba, J. Adam: a method for stochastic optimization. Preprint at *arXiv* https://doi.org/10.48550/arXiv.1412.6980 (2014).
89. Han, S., Pool, J., Tran, J., & Dally, W. Learning both weights and connections for efficient neural network. In *Proc. Advances in Neural Information Processing Systems 28* (eds Cortes, C. et al.) (NeurIPS, 2015).
90. Luo, J.-H., Wu, J. & Lin, W. ThiNet: a filter level pruning method for deep neural network compression. In *Proc. IEEE International Conference on Computer Vision* 5068–5076 (IEEE, 2017).
91. Li, H., Kadav, A., Durdanovic, I., Samet, H. & Graf, H. P. Pruning filters for efficient convnets. In *Proc. 5th International Conference on Learning Representations* 1683–1695 (ICLR, 2017).
92. Goodfellow, I. J., Shlens, J. & Szegedy, C. Explaining and harnessing adversarial examples. Preprint at *arXiv* https://doi.org/10.48550/arXiv.1412.6572 (2015).
93. Veerabadran, V. et al. Subtle adversarial image manipulations influence both human and machine perception. *Nat. Commun.* **14**, 4933 (2023).
94. Morcos, A., Raghu, M. & Bengio, S. Insights on representational similarity in neural networks with canonical correlation. In *Proc. Advances in Neural Information Processing Systems 31* (eds Bengio, S. et al.) (NeurIPS, 2018).
95. Williams, A. H., Kunz, E., Kornblith, S. & Linderman, S. Generalized shape metrics on neural representations. In *Proc. Advances in Neural Information Processing Systems 34* (eds Ranzato, M. et al.) (NeurIPS, 2021).
96. Pattisapu, S. SciDraw: monkey brain. *Zenodo* https://doi.org/10.5281/zenodo.17553661 (2025).
97. Cowley, B. R., Stan, P. L., Pillow, J. W. & Smith, M. A. Data from "Compact deep neural network models of visual cortex". *Carnegie Mellon University* https://doi.org/10.1184/R1/30500090 (2025).
98. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J. & Wojna, Z. Rethinking the inception architecture for computer vision. In *Proc. 2016 IEEE Conference on Computer Vision and Pattern Recognition* 2818–2826 (2016).
99. Pospisil, D. A., Pasupathy, A. & Bair, W. 'Artiphysiology' reveals V4-like shape tuning in a deep network trained for image classification. *eLife* **7**, e38242 (2018).
100. Olah, C. et al. Zoom in: an introduction to circuits. *Distill* **5**, e00024-001 (2020).
101. Goh, G. et al. Multimodal neurons in artificial neural networks. *Distill* **6**, e30 (2021).
102. Ponce, C. R. et al. Evolving images for visual neurons using a deep generative network reveals coding principles and neuronal preferences. *Cell* **177**, 999–1009 (2019).
103. Gallant, J. L., Connor, C. E., Rakshit, S., Lewis, J. W. & Van Essen, D. C. Neural responses to polar, hyperbolic, and cartesian gratings in area V4 of the macaque monkey. *J. Neurophysiol.* **76**, 2718–2739 (1996).
104. Pasupathy, A. & Connor, C. E. Responses to contour features in macaque area V4. *J. Neurophysiol.* **82**, 2490–2502 (1999).
105. Tanigawa, H., Lu, H. D. & Roe, A. W. Functional organization for color and orientation in macaque V4. *Nat. Neurosci.* **13**, 1542–1548 (2010).
106. Nandy, A. S., Sharpee, T. O., Reynolds, J. H. & Mitchell, J. F. The fine structure of shape tuning in area V4. *Neuron* **78**, 1102–1115 (2013).
107. Li, M., Liu, F., Juusola, M. & Tang, S. Perceptual color map in macaque visual area V4. *J. Neurosci.* **34**, 202–217 (2014).
108. Okazawa, G., Tajima, S. & Komatsu, H. Image statistics underlying natural texture selectivity of neurons in macaque V4. *Proc. Natl Acad. Sci. USA* **112**, E351–E360 (2015).
109. Lieber, J. D., Oleskiw, T. D., Simoncelli, E. P. & Movshon, J. A. Responses of neurons in macaque V4 to object and texture images. Preprint at *bioRxiv* https://doi.org/10.1101/2024.02.20.581273 (2025).
110. Carlson, E. T., Rasquinha, R. J., Zhang, K. & Connor, C. E. A sparse object coding scheme in area V4. *Curr. Biol.* **21**, 288–293 (2011).
111. Abbasi-Asl, R. et al. The DeepTune framework for modeling and characterizing neurons in visual cortex area V4. Preprint at *bioRxiv* https://doi.org/10.1101/465534 (2018).
112. Hofer, H., Carroll, J., Neitz, J., Neitz, M. & Williams, D. R. Organization of the human trichromatic cone mosaic. *J. Neurosci.* **25**, 9669–9679 (2005).
113. Brainard, D. H. Color and the cone mosaic. *Annu. Rev. Vis. Sci.* **1**, 519–546 (2015).
114. Simonyan, K., Vedaldi, A. & Zisserman, A. Deep inside convolutional networks: visualising image classification models and saliency maps. In *Proc. Workshop at 2nd International Conference on Learning Representations* (eds. Bengio, Y. & LeCun, Y.) (ICLR, 2014).

Acknowledgements We are grateful to S. Schmitt for assistance with data collection and spike sorting; to our animal care staff; and N. Rafidi for providing comments on the manuscript. We thank S. Pattisapu for the macaque illustration in Fig. 1a; we used SciDraw for the illustrations of the monkey[96] and the brain[55] in Fig. 1a. Images in experiments come from the YFCC100M dataset[65]; these original experimental stimuli have been replaced with close matches of copyright-free equivalents from Adobe Stock. Original experimental images can be found in the publicly available data repository[97]. This work was supported by a C.V. Starr Fellowship and the Pershing Square Innovation Fund to B.R.C.; a US National Institutes of Health (NIH) grant (F31EY031975) to P.L.S.; a Simons Collaboration on the Global Brain Investigator Award (SCGB AWD543027), NIH BRAIN Initiative grants (NS104899 and R01EB026946) and a U19 NIH-NINDS BRAIN Initiative Award (5U19NS104648) to J.W.P.; an NIH grant (R01EY029250) to M.A.S.; and an NIH grant (R01EY037194) to M.A.S. and B.R.C.

Author contributions B.R.C., P.L.S., J.W.P. and M.A.S. conceived of and designed the study. B.R.C. and P.L.S. designed and performed the closed-loop experiments. B.R.C. provided image stimuli. P.L.S. recorded the electrophysiological data. B.R.C. designed, trained and analysed the models. B.R.C. wrote the manuscript with input from P.L.S., J.W.P. and M.A.S.

Competing interests The authors declare no competing interests.

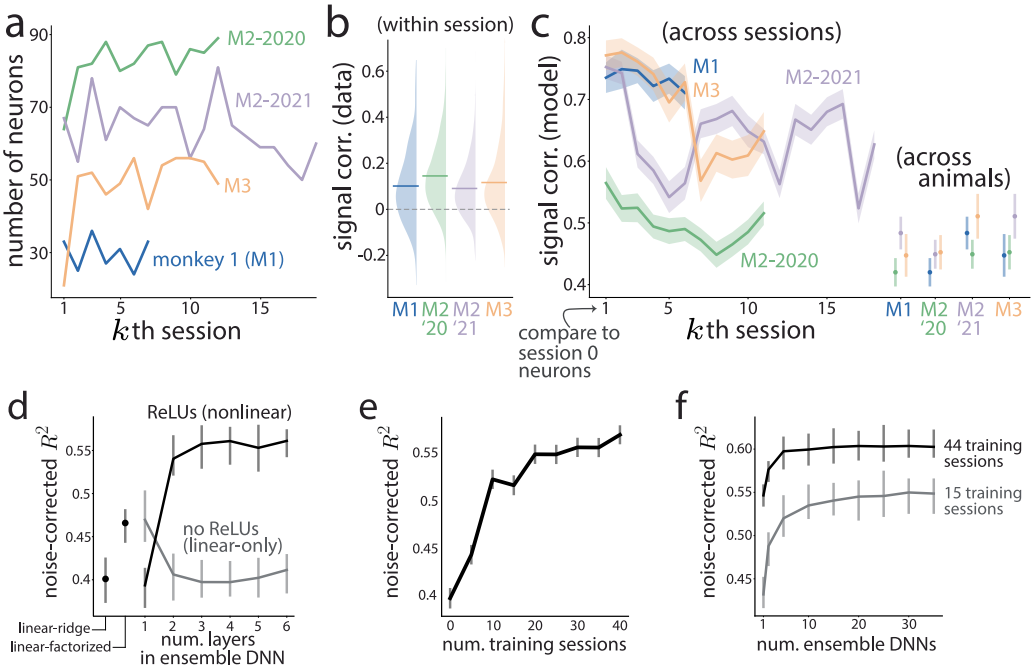
Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-026-10150-1.

Correspondence and requests for materials should be addressed to Benjamin R. Cowley or Matthew A. Smith.

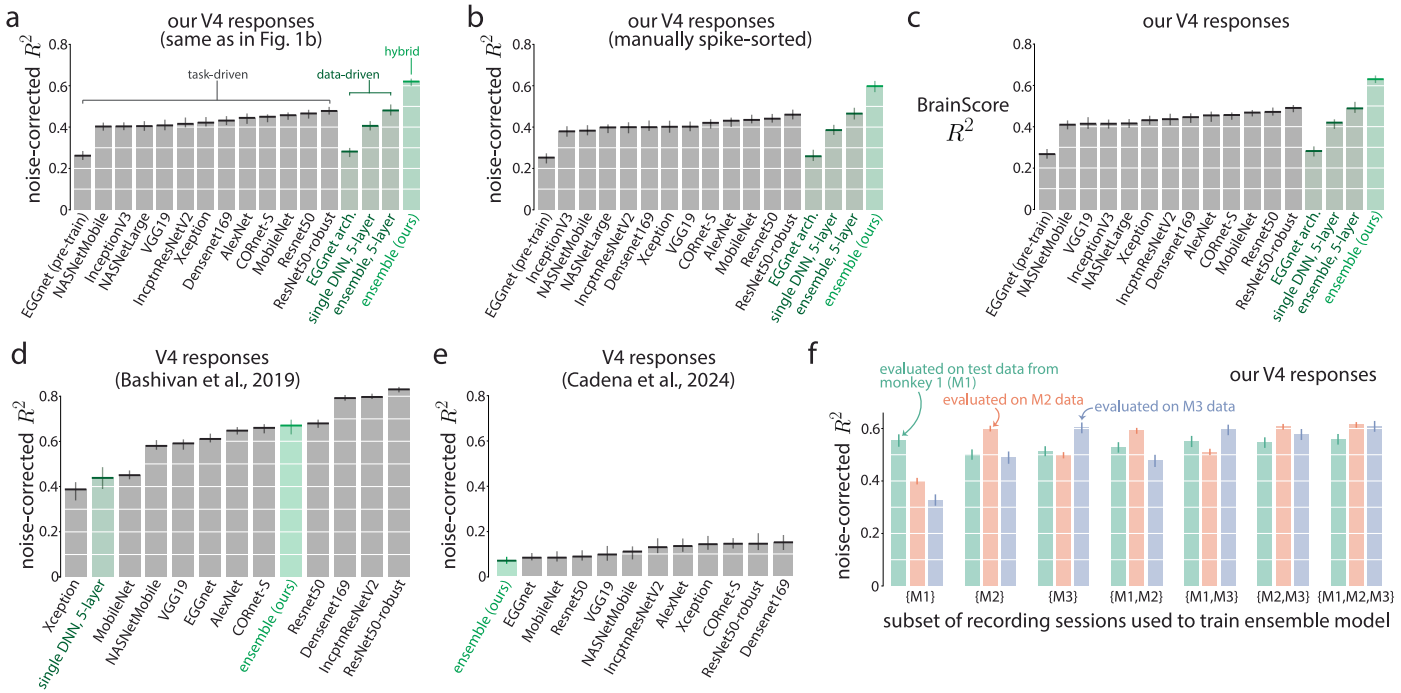
Peer review information *Nature* thanks the anonymous reviewers for their contribution to the peer review of this work.

Reprints and permissions information is available at http://www.nature.com/reprints.



Extended Data Fig. 1 | See next page for caption.

Extended Data Fig. 1 | Heterogeneity of stimulus encoding in V4 populations within and across sessions, as well as model improvements for prediction. **a**. Ideally, we would train the deep ensemble model on recorded responses of the same neurons to tens of thousands of images (each with multiple repeats), but this is difficult: Recording the same neuron in an awake animal across multiple sessions with current electrophysiological techniques, such as an electrode array, cannot be guaranteed, as the electrode array may shift between sessions or degrade in quality due to gradual build up of scar tissue. Indeed, after automated spike sorting of our V4 responses, we found that the number of neurons differed across sessions, indicating that at least subsets of neurons were not the same across sessions. Each trace is the number of neurons for one of three monkeys (M1, M2-2020, M2-2021, and M3, where M2-2020 and M2-2021 were recordings from the same monkey but 5 months apart; M1, M2, M3 correspond to identifiers WE, PE, and RA, see Methods). **b**. One possible way to overcome the small number of training images per neuron (1,000-3,000 images per session) is to leverage signal correlations across neurons within a session. In other words, if two neurons have similar stimulus tuning (for example, a preference for dots), the model may reuse its computations to predict both neurons—reducing its number of parameters and thus needing less training data. To test this, we computed the signal correlations of V4 neurons for the four test recording sessions (one per each monkey dataset). Lines indicate medians. The signal correlations between pairs of V4 neurons within a session were low (median $\rho = 0.11$ across neurons and sessions). Thus, within a session, neurons had diverse tuning preferences; we found the same result for the compact models (Fig. 3b). This motivated us to see if we could "stitch" neurons across sessions for training. **c**. One possibility to increase the number of training images per neuron is to only show one repeat per image without inter-stimulus intervals of blank grey screens[59]; however, the repeat-to-repeat variability is often too large for reliable parameter estimation (Supplementary Fig. 2). Another possibility is to repeat a small number of images each session and then match pairs of recorded units across sessions based on how correlated the pair's responses are to the images[20]. A drawback to this approach is the matching may too strongly rely on a small number of images (25 images used in ref. 20), resulting in chosen neurons all with similar stimulus tuning (see Supplementary Fig. 2). In addition, repeating the same set of images each session takes away possible presentations of unique images—presenting a set of the same 25 images would otherwise take the place of ~1,000 unique images in our experiments. To avoid these drawbacks, we sought a way to train the deep ensemble model using all available recorded units across sessions and animals. We reasoned that re-inserting an electrode array each session—that is, recording from a new set of neurons each session—would not provide enough recording time to present the number of training images needed to fit the diversity of stimulus tuning across all V4 neurons (**b**). Instead, we opted for chronically-implanted electrode arrays that would likely record from the same neurons between sessions with some neurons lost and added over time (**a**). To verify this, we computed the signal correlation between matched pairs of neurons between the first recording session (session 0) and each kth recording session. Because images differed across sessions, we used the deep ensemble predicted responses to 5,000 natural images to compute the signal correlations. For matched pairs, we started with the neuron pair with the largest signal correlation and iteratively added matches with the largest signal correlation ignoring any neurons already in a matched pair; lines indicate medians, shaded areas denote bootstrapped 90% confidence intervals ($n = 25$–89, exact corresponding values in **a**). We found that sessions nearest to the first recording ($1 \le k \le 5$) often had stable signal correlations that eventually decreased or varied as time increased between sessions. Still, the signal correlations within animals were often larger than signal correlations of pairs matched between animals (rightmost dots, where each dot denotes a median, and error bars indicate bootstrapped 90% confidence intervals over n neurons). The signal correlations between M2-2020 and M2-2021 were small (rightmost dots, 'M2'20' and 'M2'21'), indicating that the 5 months between these two recording periods were enough to record from a seemingly new set of neurons; thus, we treat these two recording periods separately. These results suggest that for the most part, recording sessions share a subset of the same neurons; in other words, the stimulus tunings of the recorded V4 population remain relatively stable across training and test sessions. We leverage this fact to train on neurons across multiple sessions—stitched together via linear mappings—and evaluate on held-out recording sessions (where neurons may differ in number or identity). **d**-**f**. Three modeling improvements that contributed to our 30% boost in prediction performance (Fig. 1b). To reduce the computational burden, we trained 5 ensemble members instead of the 25 members used for the deep ensemble model (Fig. 1b); this explains small decreases in prediction performance between Fig. 1b,d–f. In all panels, lines denote medians, error bars denote 90% bootstrapped confidence intervals, $n = 219$ neurons. First, we asked whether placing nonlinearities between ResNet50 features and V4 responses led to better prediction. We trained a 5-member deep ensemble with different numbers of skip connection layers (**d**, 'ReLUs, nonlinear'); we chose 4 skip connection layers for our final deep ensemble model (Fig. 1b). We then asked if such nonlinearity was necessary by substituting all ReLU activation functions with linear (identity) functions (**d**, 'no ReLUs, linear-only'), converting the deep ensemble members to entirely linear models. For reference, we considered prediction performance for linear mappings identified via ridge regression ('linear-ridge') or a factorized mapping ('linear-factorized', see Methods section 'Predicting V4 responses'). Thus, using a nonlinear mapping (that is, a deep ensemble with ReLU activation functions) between task-driven DNN features (from ResNet50) and V4 responses better predicted responses that using a purely linear mapping typical of most studies[10,15] or linear-only ensemble DNNs. This suggests the deep ensemble model captures important nonlinearities of V4 processing. Second, as expected, training on an increasing number of recording sessions led to better prediction (**e**). For this analysis, we varied the number of training sessions k by randomly choosing a subset of k sessions out of the 44 sessions used for training. Third, using an ensemble of DNNs versus a single DNN increased prediction performance, especially for a smaller number of training sessions (**f**). This increase is primarily because ensembling overcomes overfitting: Each DNN in the ensemble has a different random initialization and overfits differently to the small amount of training data. Averaging over the ensemble averages away differences in overfitting. For this analysis, we varied the number of ensemble DNNs by training a model with 40 ensemble DNNs and taking subsets of ensemble DNNs; for each subset, we recomputed prediction performance. To confirm that using an ensemble versus a single model is even more effective with less training data, we trained the ensembles on either all 44 recording sessions or 15 randomly-chosen recording sessions. Our final choice of 25 ensemble DNNs ensured good prediction and good estimation of ensemble disagreement (used for active learning).

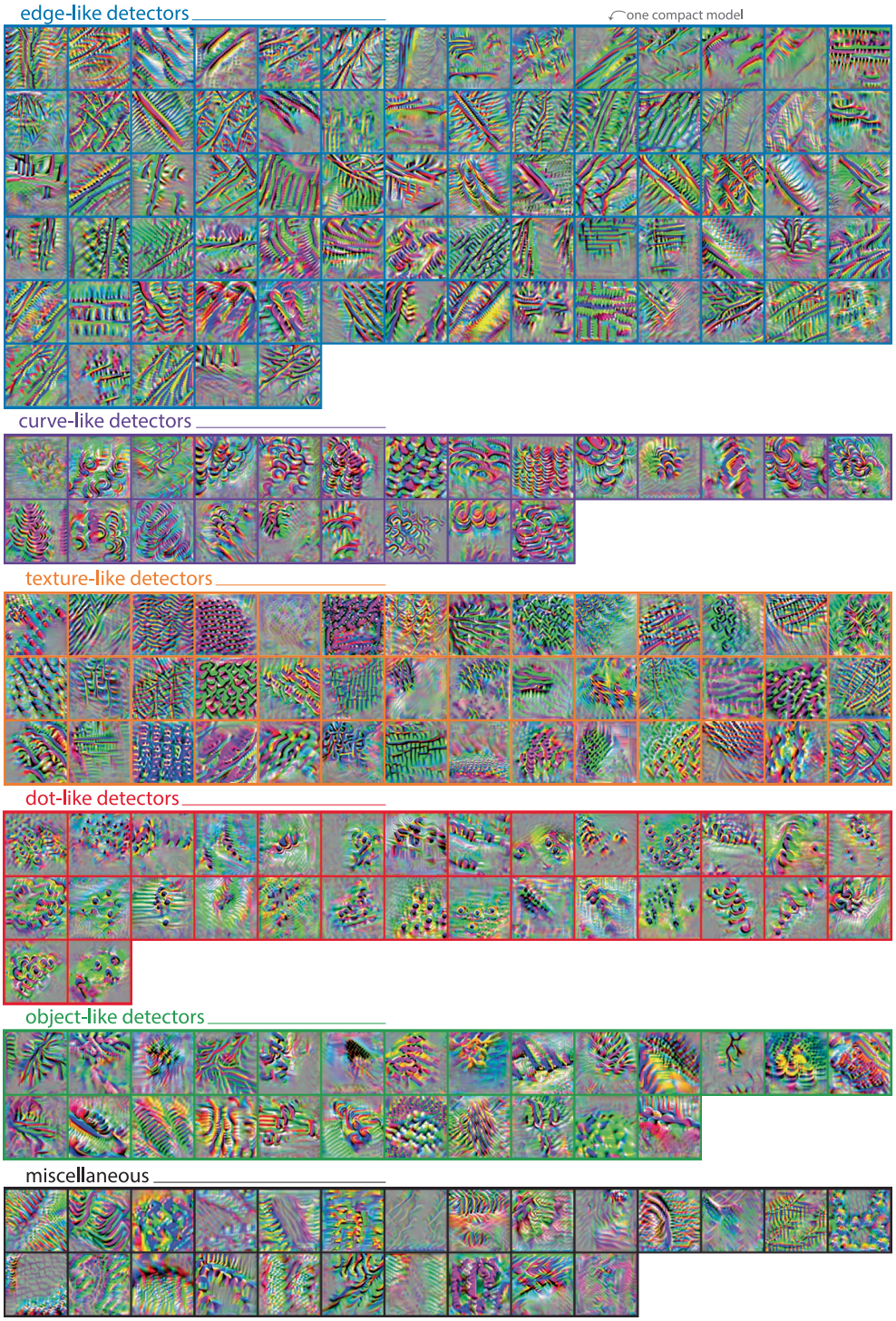


Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | Comparing our noise-corrected R^2 values versus those reported in previous studies. a. Prediction performance (noise-corrected R^2) on held-out V4 responses from our dataset for different task-driven and data-driven DNN models as well as our hybrid deep ensemble model (green). In addition to the task-driven, data-driven, and hybrid models in Fig. 1b, we also evaluated NASNetMobile[81] and InceptionV3[98], as well as a recently-proposed model architecture called EGGnet[19]. We considered two versions of EGGnet. The first is a pre-trained EGGnet ('EGGnet, pre-train'), trained to predict V4 responses from a different dataset[31]. We fixed the weights of its core network and used a factorized linear mapping to read out its embedding in the same way as for the task-driven DNNs. The poor performance of this task-driven EGGnet was not unexpected, as we also find training on one set of V4 neurons fails to generalize to other sets of V4 neurons (see panel **f**). The second version was a data-driven EGGnet trained on our 44 recording sessions in the same procedure as that for the other data-driven models. Again, we observed lower prediction performance for data-driven EGGnet than the other data-driven models. This may stem from using a factorized linear mapping versus an attention layer readout[19], especially considering EGGnet's output embedding has a much larger number of spatial dimensions (92 × 92 × 64) than our other data-driven model (28 × 28 × 100). Lines denote medians; error bars are 90% confidence intervals, $n = 219$ neurons. **b**. The V4 responses in panel **a** were spike-sorted with an automated pipeline using a pre-trained convolutional neural network for classification of spikes[57]. Here, we tested whether this automated pipeline differed too much from manual spike sorting by a human expert. We manually spike-sorted the four held-out test sessions, removing any false spikes or noisy electrode channels. We then used the deep ensemble model—trained on all other sessions whose responses were only spike-sorted by our automated pipeline—to predict these manually spike-sorted responses. Same format as in **a**, $n = 219$ neurons. We found only a small decrease in noise-corrected R^2 ($\Delta R^2 < 0.02$). This indicates that our automated spike-sorting pipeline had a reliability on par with that of an expert human annotator; this, along with our criteria for including electrode units as a "neuron" (see Methods), gave us strong confidence that our responses strongly reflected the activity of the actual recorded V4 neurons. **c**. To evaluate prediction performance, we used a newly-proposed metric R^2_{ER}[58] (mathematically defined in Methods) that is a consistent statistical estimator of the noise-corrected R^2 with expected responses (ER). We compared the R^2_{ER} metric to the commonly-used BrainScore R^2 metric, which computes an R^2 ceiling by splitting repeats into two halves, corrected with the Spearman-Brown procedure[15]. The BrainScore R^2 is computed with the following pseudocode:

$\mathbf{R} : (N_{\text{images}} \times N_{\text{repeats}})$ `// matrix of V4 responses over repeats`
$\bar{r} = \text{row mean}(\mathbf{R}) : (N_{\text{images}} \times 1)$ `// mean over repeats for each row`
$\hat{r} : (N_{\text{images}} \times 1)$ `// model predictions`
$\rho_{\text{model_vs_data}} = \text{corr}(\bar{r}, \hat{r})$ `// correlation over images`
for $i_{\text{run}} = 1$ **to** N_{runs} **do**
  shuffle repeats per row in $\mathbf{R}$
  $\bar{r}_{\text{split1}} = \text{row mean}(\mathbf{R}[:, 1 \text{ to } N_{\text{repeats}}/2])$
  $\bar{r}_{\text{split2}} = \text{row mean}(\mathbf{R}[:, N_{\text{repeats}}/2 + 1 \text{ to } N_{\text{repeats}}])$
  $\rho_{\text{split}}[i_{\text{run}}] = \text{corr}(\bar{r}_{\text{split1}}, \bar{r}_{\text{split2}})$
end
$\bar{\rho}_{\text{split}} = \text{mean}(\rho_{\text{split}})$ over runs
$\rho_{\text{ceiling}} = 2\bar{\rho}_{\text{split}}/(1 + \bar{\rho}_{\text{split}})$ `// Spearman-Brown correction`
$\rho_{\text{model_vs_model}} = \text{corr}(\hat{r}, \hat{r}) = 1$ `// equals 1 as model is deterministic`
BrainScore $R^2 = \rho^2_{\text{model_vs_data}}/(\rho_{\text{model_vs_model}} \cdot \rho_{\text{ceiling}})$

While both R^2 estimators should converge on the same R^2 value given enough data[58], it was unclear whether these estimators differed for our V4 dataset with its numbers of images and repeats. We re-evaluated the deep ensemble model predictions with the BrainScore R^2 metric and found little difference between the two estimators (compare with results in **a**; same format, $n = 219$ neurons). This indicates that the BrainScore R^2 metric and noise-corrected R^2_{ER} we use are mostly identical for our V4 data, suggesting our number of repeats (<10) are sufficient for estimating noise-corrected R^2. **d**. A recent study reported a BrainScore $R^2 = 0.89$ using ResNet50-robust to predict V4 responses[20], substantially larger than our reported noise-corrected $R^2 = 0.48$ for ResNet50-robust and $R^2 = 0.62$ for the deep ensemble model for our data (**a**) as well as the reported BrainScore $R^2 \approx 0.6$ for the top-performing task-driven DNNs on the BrainScore leaderboard for V4 data[15]. Using the Bashivan et al. V4 data ($n = 131$ neurons), we largely reproduced their R^2 results (same format as in **a**), where ResNet50-robust was the most predictive model with noise-corrected $R^2 = 0.83$. Because this V4 dataset only had enough images per neuron to train a factorized linear mapping (and not the shared weights of the deep ensemble), our deep ensemble model's prediction performance was on par with its ResNet50 backbone DNN ('ensemble' and 'ResNet50'). We suspect that with more training sessions, the deep ensemble model's performance would outperform task-driven DNN models, as we see in our V4 data (**a**). An open question is how many V4 neurons would be needed to train a data-driven foundation model, such as our deep ensemble model, until this model could reliably predict out-of-the-box a newly recorded V4 neuron with a single recording session. **e**. Another recent study reported performance of ResNet50-robust predicting V4 responses as a test correlation $\rho = 0.4$ and a fraction explainable variance explained FEVE = 0.12; FEVE is similar to computing our noise-corrected R^2 as well as the BrainScore R^2. We predicted this study's V4 responses ($n = 255$ neurons) and, consistent to their reported FEVE values, also found a low noise-corrected $R^2 = 0.15$ (same format as in **a**). This, along with the results in **d**, suggest that noise-corrected R^2 may considerably vary across studies, making within-study comparisons more important. We investigate the reasons for these differences in R^2 in Supplementary Fig. 2. The deep ensemble model (trained on our V4 data and fit with a factorized linear mapping between its embeddings and V4 responses) was outperformed by the task- and data-driven DNN models ('ensemble' furthest left) but had close performance to its ResNet50 backbone DNN ('ResNet50' also had $R^2 < 0.1$). A previous study[19] found EGGnet, trained on this same V4 data, to outperform ResNet50-robust; here, we find the opposite ('EGGnet' left of 'ResNet50-robust'). This is likely for two reasons. First, we use a factorized linear mapping to fit DNN features to neural responses instead of an attention layer (used by ref. 19); because the output embedding of EGGnet has a high-dimensionality (92 pixels × 92 pixels × 96 channels), the linear mapping has a large number of parameters to fit. Second, we use a pre-trained version of EGGnet and thus there may be overfitting to the same training data. Given the differences in reported noise-corrected R^2 values in our work (**a**), in Bashivan et al. (**d**), and in Cadena et al. (**e**), we further investigated the reasons for these differences (see Supplementary Fig. 2 for detailed analyses). We found that these R^2 differences primarily arise from the number of repeats for training images, the size of the neurons' spatial receptive fields relative to the presented images, and how similar the tuning is across neurons. **f**. Given the better prediction performance of the deep ensemble model relative to other models on our own V4 data versus that of V4 data from other studies (**a** versus **d** and **e**), we wondered if this could be explained by the fact that the ensemble DNNs were only trained on our V4 data. To test this, we trained the deep ensemble model on our V4 data from different subsets of animals and then evaluated the model on responses from individual held-out test sessions. Each bar represents the median R^2 across neurons from monkey 1 (M1), monkey 2 (M2), or monkey 3 (M3), corresponding to identifiers WE, PE, and RA; error bars denote bootstrapped 90% confidence intervals, $n = 33, 89, 42$ neurons for M1, M2, M3, respectively. As expected, training on data from a single animal best predicted its corresponding test data ({M1}, {M2}, {M3}). Training on data from two animals did not improve prediction performance for the test data from the left-out animal (for instance, evaluating M3 test data for training on {M1,M2}). These results suggest that to achieve an out-of-box performance better than that of a task-driven DNN, data-driven foundation DNN models will need to train on sessions from a large number of animals (or many different V4 neurons). For predicting responses from a targeted set of recorded V4 neurons, one may rely on a data-driven approach when prediction is most important (as in our work to identify the underlying computations), whereas relying on a task-driven approach may be more appropriate when recording time is limited.

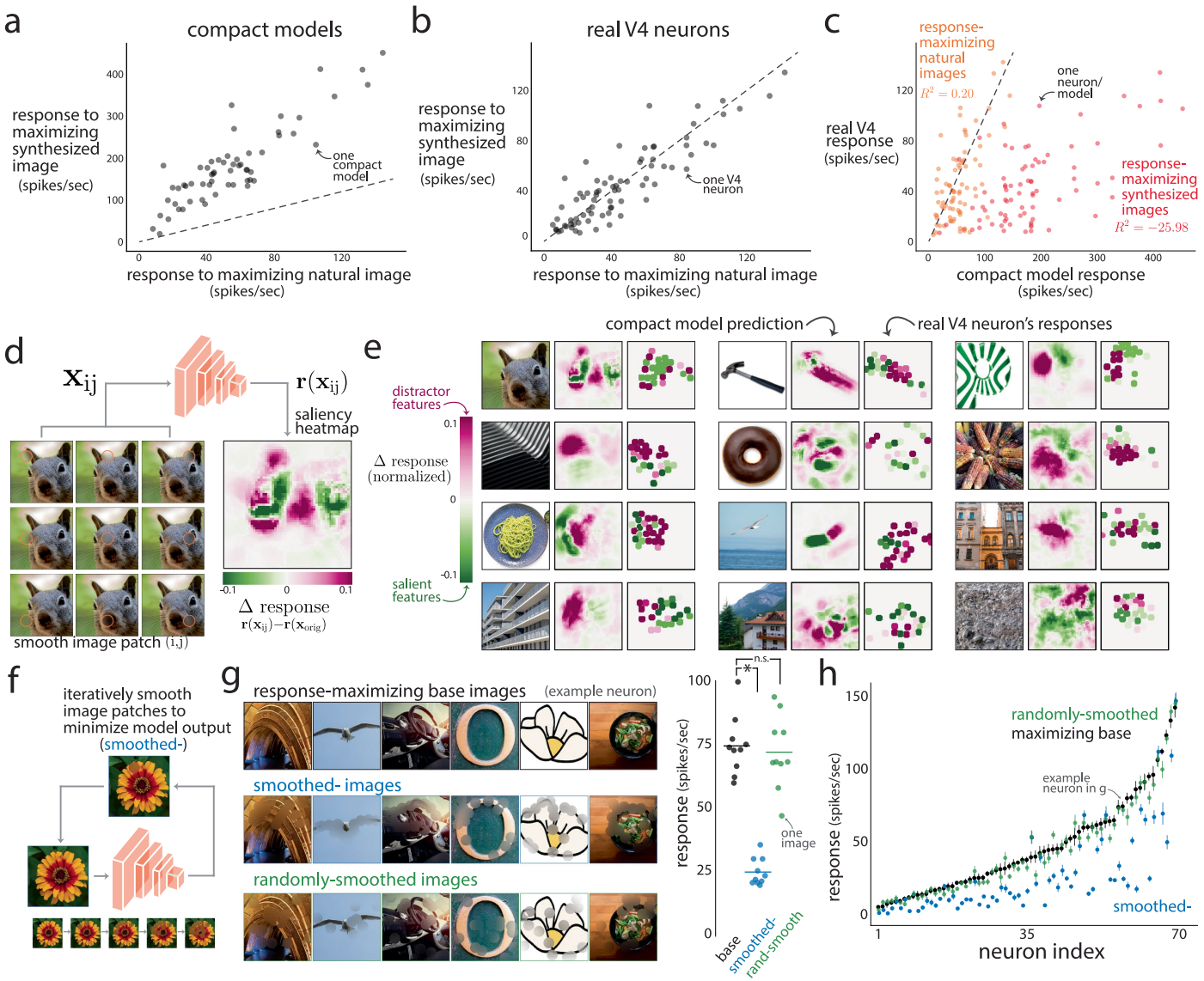


Extended Data Fig. 3 | See next page for caption.

Extended Data Fig. 3 | Response-maximizing synthesized images for all compact models. For each compact model, we computed its response-maximizing synthesized image via gradient ascent techniques (Fig. 2a). Here, we show these images for all compact models (one image per compact model). For visual clarity, we loosely grouped images into categories by eye. We make no claims about grouping of V4 neurons; in fact, the mean signal correlation squared across all pairs of compact models was low ($\rho^2 = 0.11$ between model responses to 10,000 natural images, see Fig. 3b, 'output'). This remained true when controlling for spatial receptive field location and size (Fig. 3b, 'layer 5'). Thus, the stimulus preferences across V4 neurons appear to be largely heterogeneous, allowing for a highly-expressive set of features for downstream processing in IT and other brain areas[36,43,99–102]. The image statistics of the preferred images of V4 neurons—edges, curves, textures, dots, etc.—largely match those observed in many previous studies[9,27,30,103–109], including those that synthesize preferred images[19,20,31,110,111]. Our work provides three novelties: We consider response-maximizing synthesized images in color (which, to our knowledge, have not yet been identified for V4); we explore a large number of natural images (500,000 images) to find the natural images that maximize a neuron's response (Fig. 2b); and we identify slight perturbations of natural images that elicit large changes in responses (ε-perturbed images, see Fig. 2d–f). A salient observation is the large extent to which greens, magentas, and yellows appear in our synthesized preferred images. Many V4 neurons have strong preferences for color[27,105,107]; a preference for greens and magentas versus blues may arise from the fact that primate retina has many more L and M cones (that detect red and green hues) compared to S cones (that detect blue hues)[112,113]. An exciting future direction is to characterize the strength of color processing in V4 and how it relates to natural image statistics. Overall, our response-maximizing synthesized images (MSIs) (also known as most exciting inputs) are largely in agreement but perhaps not as crisp or object-detailed as in other studies[19,20,31]. Our MSIs have shape 112 pixels × 112 pixels (versus the typically displayed 224 pixels × 224 pixels MSIs) and are in color (versus greyscale that previous studies use), which may explain the differences in crispness. It is reassuring that the MSIs of our 5-layer compact networks have similar statistics to those of MSIs derived from large task-driven DNNs, as MSIs from the latter may more strongly reflect the internal image statistics priors of task-driven DNNs versus the true preferences of V4 neurons. For example, a V4 neuron may truly prefer two dots but the task-driven DNN may embellish its MSI with two eyes (likely the most common occurrence of two dots in a large image dataset). We make no claims that our MSI approach is the best, but rather use the MSI approach to experimentally validate the predicted stimulus preferences of the compact models. A head-to-head comparison is needed to determine which type of MSIs drive V4 responses the strongest, which should also include the response-maximizing natural images chosen from a large candidate pool (500,000 images or greater).

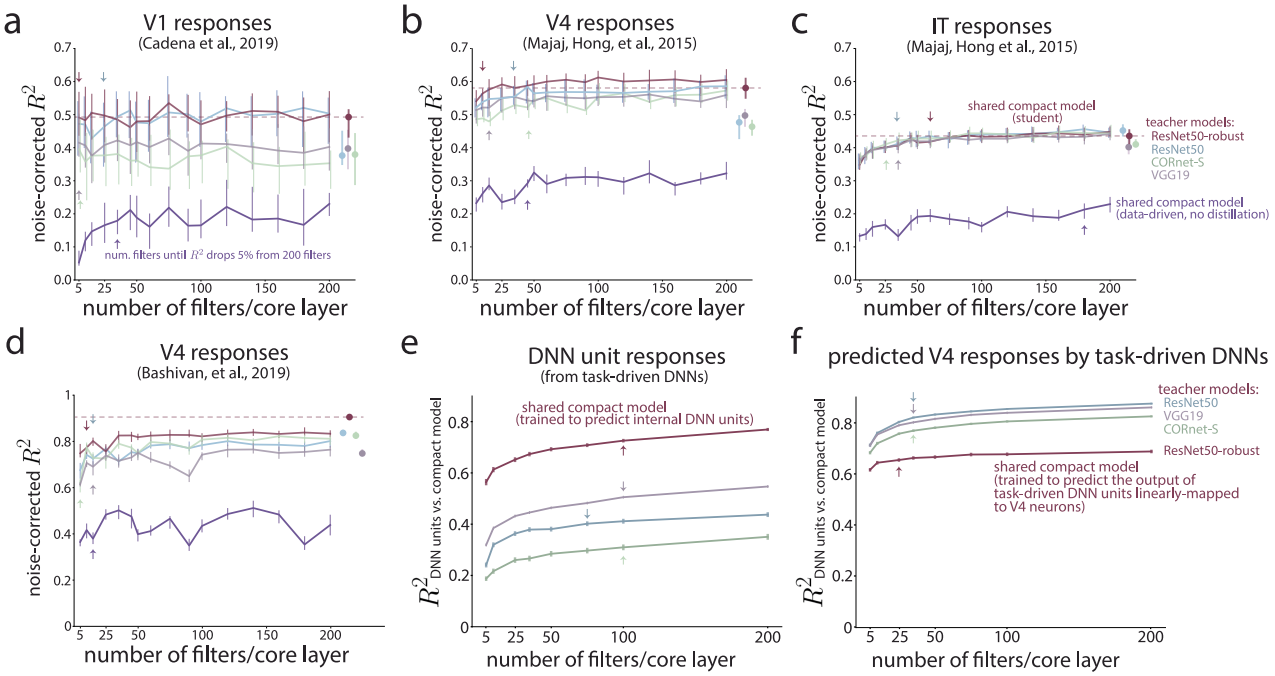
Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | V4 responses to maximizing images. The response-maximizing images of the compact models drove larger responses compared to those of randomly-chosen natural images (Fig. 2b,c). Here, we take a closer look at all responses to these maximizing images as well as compare the statistics between the response-maximizing natural images, response-maximizing synthesized images, and the natural images with the largest responses. **a**. In our validation experiments, we matched each compact model with a corresponding V4 neuron and tested whether the maximizing images predicted by the compact models strongly drove V4 responses. One possibility is that the compact models identified maximizing images that strongly drove all V4 neurons together without specificity for each V4 neuron's stimulus preference—for example, an image with high contrast likely evokes larger responses for all neurons than an image with low contrast. To see if this were the case, we examined the repeat-averaged V4 responses (left heatmap for monkey 3, RA, and right heatmap for monkey 2, PE) to response-maximizing natural images, response-maximizing synthesized images, and randomly-chosen natural images from our validation experiments (Fig. 2b,c). For each V4 neuron, we normalized its responses between 0 and 1 and sorted neurons (rows) based on the average normalized response to the natural images. For monkey 3 (left heatmap), we trained 42 compact models on data from previous recording sessions and probed the compact models for their response-maximizing natural and synthesized images. We then presented these probing images in a following session, matching neuron to compact model via the randomly-chosen natural images (see Methods). We organized the columns for the response-maximizing natural images such that the first 10 columns were responses to the maximizing natural images predicted by the first compact model; the next 10 columns were for the second compact model, and so on; the response-maximizing synthesized images were ordered in the same way. For the natural images, we sorted columns based on the mean response across all V4 neurons, where the largest mean response was the rightmost column. We organized the response heatmap for monkey 2 (right heatmap) in a similar way; for this animal, only one response-maximizing natural image and one response-maximizing synthesized image were generated for each of 50 compact models. We found that it was not the case that the response-maximizing images for one compact model drove the entire V4 population (red squares in columns are sparse) and that few neurons were activated by all response-maximizing images (red squares in rows are sparse). We also found similar response patterns to response-maximizing natural images and response-maximizing synthesized images (left heatmap, compare heatmaps between natural images and synthesized images; this was more difficult to see in the right heatmap due to one image per compact model), consistent with both sets of response-maximizing images having similar image statistics. Our main finding was that the response-maximizing images—predicted by the compact models trained only on previous sessions—yielded larger responses than responses to randomly-chosen natural images (compare left to right in each heatmap, also see Fig. 2b,c). **b**. We noticed that some of the responses to randomly-chosen natural images were larger than the response-maximizing images predicted by the compact models (black dots, 'natural images with largest responses'); each dot denotes the repeat-averaged response of one V4 neuron to one image. Because the randomly-chosen natural images could include images similar to the response-maximizing images, we wondered to what extent these natural images with largest responses resembled the response-maximizing images. For 8 example neurons, we found close matches in image statistics between response-maximizing natural images (orange), response-maximizing synthesized images (red), and natural images with largest responses (black). For example, V4 neuron 1 preferred images with buttons and circles as well as images with round human faces and curved objects. V4 neuron 2 preferred images with a vertical object with a high-spatial frequency texture. V4 neuron 3 preferred images with dots as well as images with eyes and far away objects. This further increased our confidence that the preferred stimuli as predicted by the compact models were good approximations for the true preferred stimuli of the real V4 neurons. Images were reproduced from Adobe Stock.

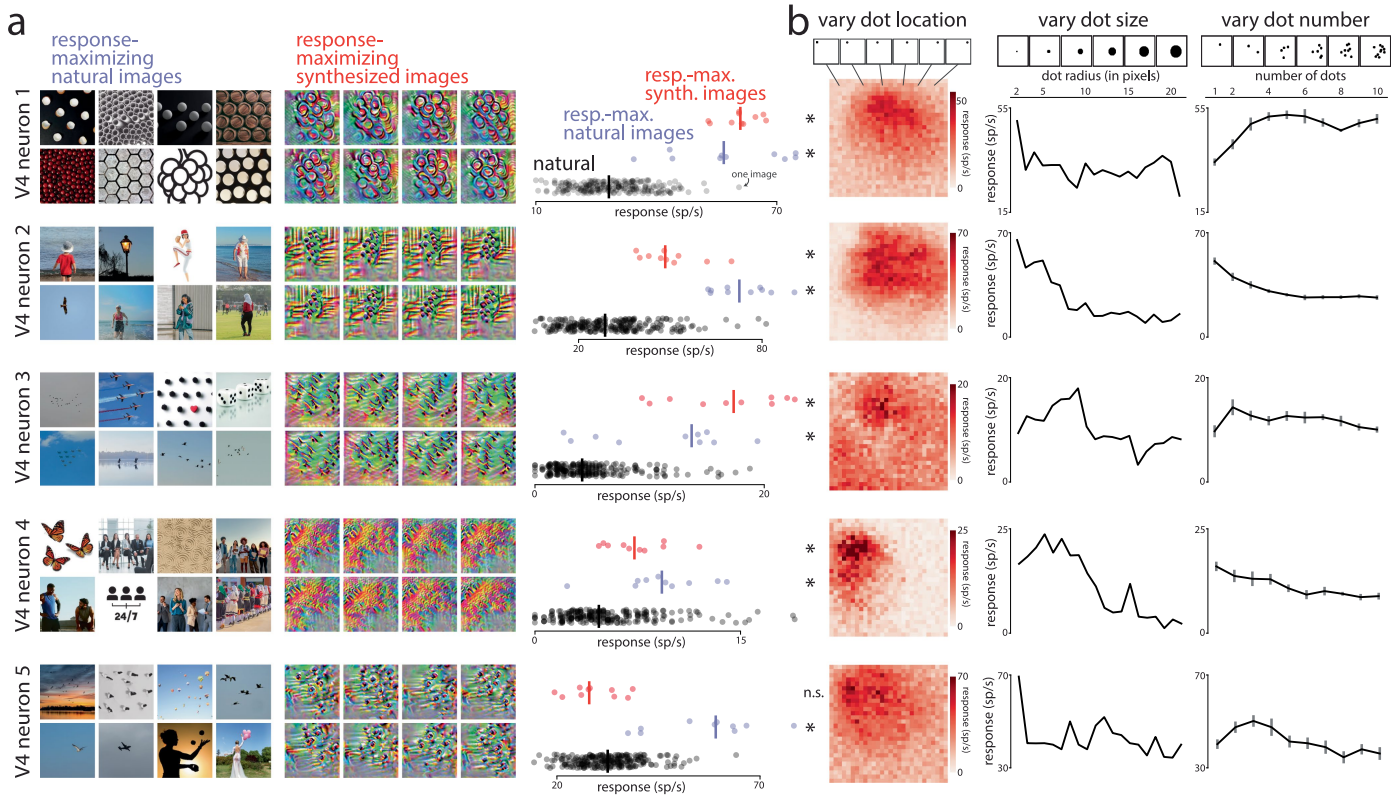


Extended Data Fig. 5 | See next page for caption.

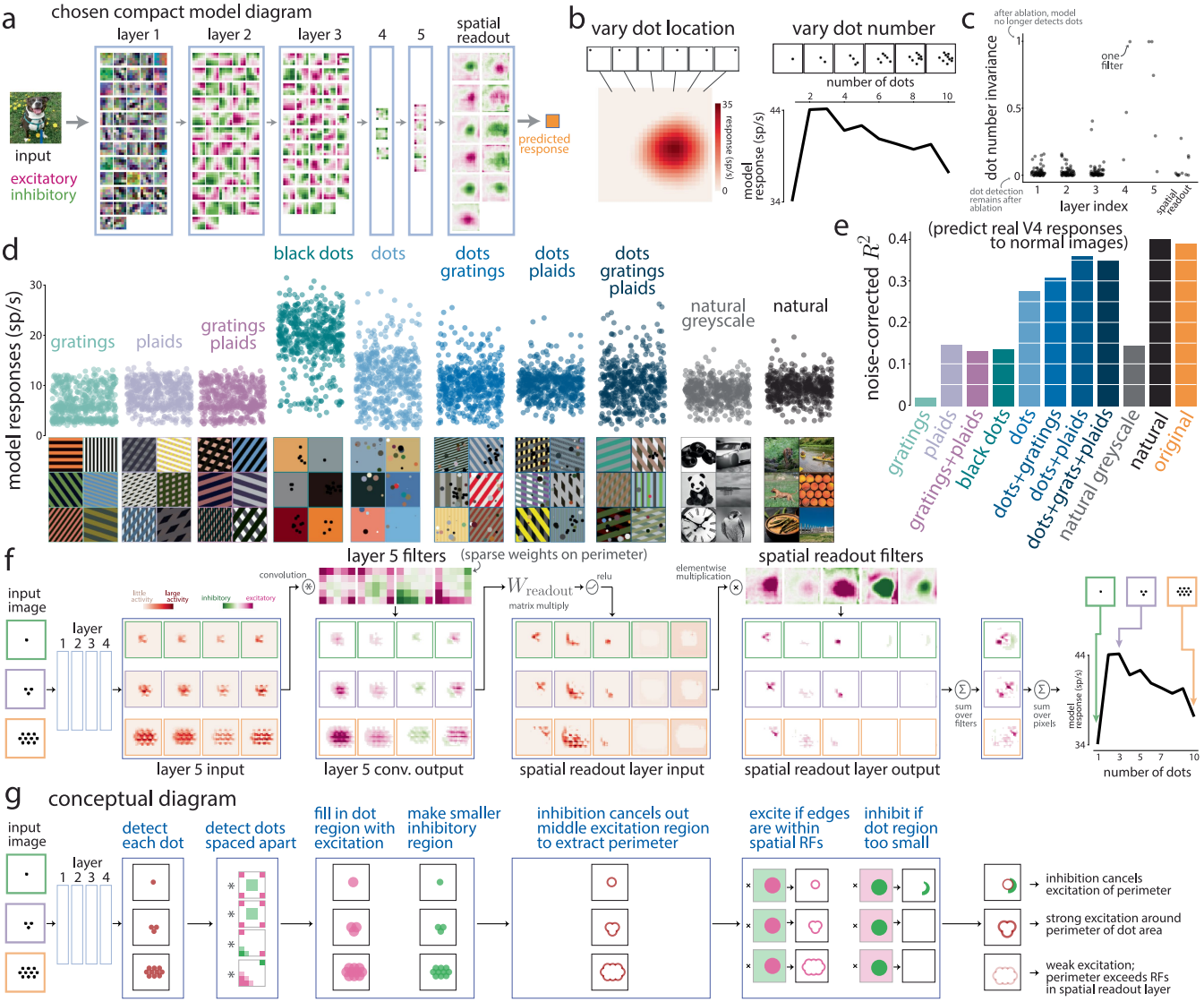
Extended Data Fig. 5 | Further experimental validations of the compact models. Our compact models' predictions held up to experimental validation (Fig. 2), including identifying response-maximizing natural and synthesized images. Here, we compare which type of these images better drives V4 responses. In addition, we present results of further experimental validations, including identifying saliency maps and "ε-perturbed smooth" images. **a**-**c**. We found that the compact models predicted much stronger V4 responses to response-maximizing synthesized images versus response-maximizing natural images (**a**, dots above dashed line; difference of means between maximizing synthesized and maximizing natural: 111.7 spikes/sec, $p < 0.002$, permutation test, $n = 78$). However, this was not the case for the real responses—both types of images evoked similarly large responses (**b**, dots hug dashed line, difference of means not significant, $p = 0.836$, permutation test, $n = 78$). This difference largely stems from the compact models' inability to predict V4 responses to maximizing synthesized images (**c**, red dots, coefficient of determination $R^2 = -26.0$, not noise-corrected) whereas their prediction for maximizing natural images remains relatively intact (**c**, orange dots, coefficient of determination $R^2 = 0.2$, not noise-corrected). Each dot denotes the response of a single neuron, averaged over repeats and maximizing images (if multiple maximizing images were shown for a V4 neuron, see Fig. 2b,c). This inability to predict maximizing synthesized images was not unexpected—the optimizing procedure had full access to every weight and every pixel to optimize an image tailored to that compact model, and some "adversarial" noise was expected[32,34]. Moreover, the resulting synthesized images were well outside the distribution of training images, and we would expect poor prediction for these outlying regions of image space. This was one motivation for training our deep ensemble model with closed-loop active learning (Fig. 1c) in which we trained the model on out-of-distribution images. We were also surprised that the maximizing natural images evoked V4 responses as large as those to maximizing synthesized images (**b**). This suggests that one cannot rule out choosing from a large pool of candidate images (in this case, 500,000 candidate images) to maximally drive V4 responses. **d**. A commonly-used approach to explain a DNN's output is to identify which parts of an input image are the most relevant or "salient" for the DNN's prediction; this approach is called saliency analysis[114]. One implementation is to smooth a small patch of the image (small orange circles in example images denote smoothed patches; these circles were not present in actual stimuli) and see if the resulting response is larger or smaller than the response to the original image. An increase in response (pink) indicates that the visual feature within the smoothed patch is distracting, as removing this feature leads to a *larger* response. Likewise, a decrease in response (green) indicates a salient or excitatory visual feature. We fed as input a set of images where the (i,j)th image had a smoothed image patch centered at (i,j). We then formed a heatmap of the resulting responses $\mathbf{r}$ based on the (i,j)th image patch (forming a 'saliency heatmap'). For this example image of a squirrel and the chosen compact model, the most salient features are the eyes (green regions), while the most distracting features are the fur texture and the edges around the left eye (pink regions). The squirrel images were reproduced from Adobe Stock. **e**. In our validation experiments, we probed the trained compact models to identify maximizing natural images (Fig. 2b). For each maximizing natural image, we computed the saliency heatmap of the compact model ('compact model prediction') following the procedure in **d**. We then used these predictions to smooth 25 non-overlapping image patches that led to the largest changes in responses (where the number 25 was chosen as a compromise between covering as much as the image as possible within recording time constraints). In a following session, we showed each 'base' image as well as the 25 smoothed images, each with one smoothed image patch. For each example image shown here, we matched a V4 neuron with the image's corresponding compact model (in the same way as in Fig. 2b,c) and computed the resulting saliency heatmap for V4 neurons (rightmost panels). Responses were z-scored using the mean and standard deviation estimated with the V4 neuron's responses to all natural images shown in the session. We found that V4 neurons did vary their responses to local smoothing and that these changes in responses largely matched those predicted by the compact models. Thus, for a given image, a V4 neuron's response can be suppressed and excited by different local visual features; our compact models can be used to predict which features at which locations. The natural images were reproduced from Adobe Stock. **f**. Inspired by the saliency approach in **d** and **e**, we experimentally validated our compact models by having them predict which visual features of an image to smooth in order to minimize a V4 neuron's response. We first began with the compact model's maximizing natural image as a base image. Then, in a greedy manner, we iteratively chose an image patch to smooth that led to the most suppressed response as predicted by the compact model ('smooth-', see orange circle). Successive iterations added image patches that did not overlap with previously-chosen image patches. This led to an image with specific visual features smoothed away. Bottom inset: a sequence of images for which a base image is cumulatively smoothed at different patches determined by the compact model's predictions; the final smoothed image is the rightmost. Images were reproduced from Adobe Stock. **g**. Example base images (left, top row, 'maximizing base images'), smoothed versions to minimize the model output response as predicted by a compact model (left, middle row, 'smoothed- images'), and images for which randomly-chosen patches were smoothed as a control (left, bottom row, 'randomly-smoothed images'). The randomly-smoothed and smoothed- images had the same number of pixels smoothed. These example maximizing base images and randomly-smoothed images elicited similarly large responses from a V4 neuron (right, black versus green dots, $p = 0.70$, two-sided permutation test, $n = 10$) whereas the smoothed- image led to a substantially smaller response (black versus blue dots, $p < 0.001$, permutation test, $n = 10$, significance denoted by an asterisk). Each dot denotes the repeat-averaged response to one image. The response-maximizing base images were reproduced from Adobe Stock. The smoothed- and randomly smoothed images were adapted from Adobe Stock. **h**. Responses for all V4 neurons from two recording sessions (each session from a different animal). V4 neurons were matched to compact models via held-out images (same procedure as used in Fig. 2b–f). For one session, only one base image was shown per compact model; for these images, dots denote repeat-averaged responses with no error bars. For the other session, we presented 10 base images (and their smoothed- and randomly-smoothed counterparts) per neuron. For this session, dots denote the average response over the 10 images, and error bars denote 1 s.e.m. of the repeat-averaged responses over the 10 images. Neurons were sorted based on mean response to the base images. We found that responses to smoothed-images were roughly half as small as responses to the base images across V4 neurons (blue versus black dots, normalized percent change computed as $\%\Delta\bar{r} = 100 \cdot [\bar{r}(\text{smoothed}-) - \bar{r}(\text{base})]/\bar{r}(\text{base})$: mean $\%\Delta\bar{r} \pm$ s.e.: $-46.1\% \pm 3.5\%$, where $\bar{r}$ denotes a neuron's repeat-averaged response). There was little to no decrease between randomly-smoothed and base images (green versus black dots, mean $\%\Delta\bar{r} \pm$ s.e.: $-3.7\% \pm 1.9\%$). Thus, confirmed via experimental validation, the compact models accurately predicted which visual features (and their spatial information) were most salient to the V4 neurons. This provides further evidence that the compact models accurately capture the stimulus preferences of V4 neurons—and what visual features in those preferred stimuli are most important to the V4 neuron.



Extended Data Fig. 6 | Compressing models of V1, V4, and IT neurons as well as DNN units. a-c. Given the large amount of compression observed for shared compact models trained on the predicted responses of task-driven DNN models for our V4 response dataset (Fig. 3c), we wondered to what extent we could compress these models on different response datasets from brain areas along the visual stream. To this end, we compressed ResNet50-robust predicting macaque V1, V4, and IT datasets (Fig. 3d–f); here, we measure the compressibility of three other task-driven DNNs: ResNet50, CORnet-S, and VGG19 (the data-driven shared compact models, purple traces, are the same as in Fig. 3d–f). We trained shared compact models via distillation, varying the number of filters in the first three "core" layers (see Methods). We then compared the noise-corrected R^2 of these shared compact models with the full prediction performance of their task-driven DNN counterparts (rightmost dots). Interestingly, some compact models outperformed the full task-driven DNNs (**b**, blue, green, mauve traces above corresponding dots), sometimes observed with knowledge distillation[16]. Overall, the general trends of ResNet50-robust hold for all of other task-driven DNNs: Task-driven DNNs of cortical neurons are highly compressible, with the most compression achieved for V1, followed by V4, and then IT. We verified that these compact models achieve similar prediction as large task-driven DNNs but are orders-of-magnitude smaller by submitting the compact models to the public benchmark BrainScore; the compact models are now state-of-the-art predicting the corresponding V4 and IT datasets (Supplementary Table 2). Traces and dots denote medians; error bars denote bootstrapped 90% confidence intervals, $n = 115, 88, 168$ neurons for **a**, **b**, **c**, respectively. Arrows denote the smallest number of filters needed to achieve a prediction performance no less than 5% of that for 200 filters/core layer. **d.** We also evaluated the compressibility of the task-driven DNNs on the V4 response dataset from Bashivan et al.[20] ($n = 131$ neurons), which we analyzed in comparison to our own V4 response dataset (Extended Data Fig. 2). Similar to all other datasets (Fig. 3c–f), we found high compressibility (arrows on 10-15 filters). These results provide further evidence that task-driven DNN models of V4 neurons can be greatly reduced in size by orders of magnitude. **e.** Given the compressibility of models of V4 neurons, we wondered to what extent we could compress a model of responses from DNN units (that is, internal units from an intermediate layer of a task-driven DNN), treating these units as V4 neurons. To test this, we trained shared compact models to predict responses of 219 DNN internal units (whose number equaled that of the number of our recorded V4 neurons in Fig. 3c). DNN units were chosen as the hidden units centrally located in the activity map (that is, for 14 × 14 pixel activity maps, we chose hidden units with spatial locations [7,7]) with the largest activity variance over 5,000 natural images; layers matched those most predictive of our V4 responses (see Methods). We trained these shared compact models (same architecture as those in **a**–**d**) to predict DNN unit responses taken directly from a task-driven DNN to 12 million images and varied the number of filters in each of the "core" layers. We computed the raw R^2 between DNN unit responses and the predicted responses of the shared compact models; because the DNN units were deterministic, we did not need a noise-corrected R^2 metric. Despite there being no noise in the DNN unit responses, the 200-filter shared compact models failed to achieve good prediction performance for most task-driven DNNs ('200', traces below $R^2 = 0.6$; traces indicate median R^2s and error bars denote 90% confidence intervals over 219 units). Beyond this, the DNN units were not easily compressible (arrows at 75-100 filters). That an individual DNN unit is not compressible likely arises from the fact these DNN units must carry out complex computations but must read out from only ~1,000 channels from an upstream layer to aid in performing object recognition; V4 comprises hundreds of millions of neurons and may not require each individual V4 neuron to carry out many computations on its own. **f.** We wondered whether the DNN units were truly more compressible than V4 neurons or that they were less compressible because our R^2 metric did not include a noise-ceiling; in other words, the DNN unit responses had spurious nonlinearities that would otherwise be hidden by a noise-corrected R^2. To see if this were the case, we first fit factorized linear mappings from the activity of task-driven DNNs (same layers and including the same DNN units as in **e**) to the 219 V4 neurons from the 4 test recording sessions. In other words, this setting differs from **e** only with a linear mapping appended to all DNN units from that layer of the task-driven DNN. We then distilled this model with shared compact models, following the same procedure as in **e**. In contrast to the DNN units, we observed strong compressibility: The shared compact models with 200-filters have good prediction (rightmost end of traces, all traces have $R^2 > 0.6$), strong plateauing for larger numbers of filters (for example, between 50 and 200 filters/core layer), and only a small number of filters were needed to be within a 5% drop from the performance of 200 filters/core layer (arrows), consistent with the V4 results across datasets (**b** and **d**, as well as Fig. 3c). These results rule out the possibility that DNN units fail to compress because of small, spurious nonlinearities. Instead, it appears that linear readouts from DNN units that align with V4 neurons are more compressible than the DNN units themselves. An exciting possibility is that V4 may comprise expressive yet simple functions to ensure accurate stimulus encoding with robustness to noise, and that task-driven DNNs have only partially arrived at these compressible representations (that is, they are still susceptible to adversarial noise, etc.). By optimizing linear readouts of DNN units to achieve high levels of compression, one may achieve a task-driven representation more similar to that of V4. Future experiments are needed to ensure subtle nonlinearities of V4 neurons are not hidden behind repeat-to-repeat variability.

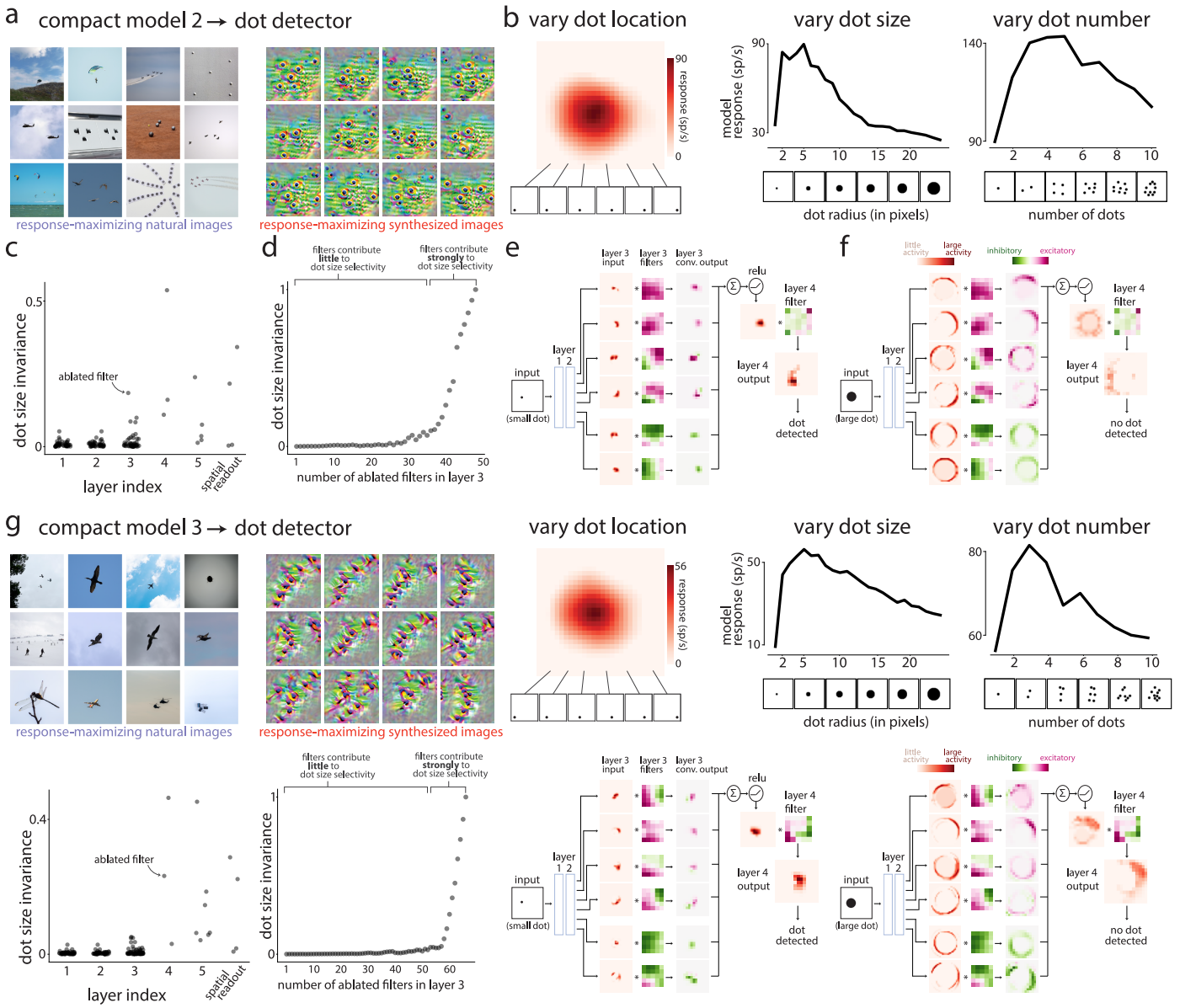


Extended Data Fig. 7 | Identifying real V4 dot detectors with experimental validation. To confirm the presence of dot-detecting V4 neurons, we ran validation experiments specifically tailored for compact models that resemble dot detectors. We first identified compact models by training on previous recording sessions. From the identified compact models, we chose 5 compact models that most resembled dot detectors based on their stimulus preferences (response-maximizing natural and synthesized images) and their responses to artificial dot stimuli (see Fig. 4a,b and Extended Data Fig. 8). We note that the chosen dot detecting compact model in Fig. 4 was not one of the chosen, as this compact model matched to a V4 neuron from another animal. For a future recording session, we presented the response-maximizing natural and synthesized images of the five chosen models as well as artificial dot stimuli (same dot stimuli as in Fig. 4b and Extended Data Fig. 8b). We identified the five recorded V4 neurons that best matched the predictions of the five compact models (by computing the noise-corrected R^2 from all other images shown in the session, same procedure as in Fig. 2b–f) and show their responses here. **a**. The response-maximizing natural images (left, examples) and response-maximizing synthesized images (middle, examples) chosen from the five compact models tended to more strongly drive responses than responses to randomly-chosen natural images (right, 'resp.-max. synth. images' and 'resp.-max. natural images' dots more to the right than 'natural' black dots). Each dot is the repeat-averaged response to one image; lines denote medians. All response-maximizing stimuli yielded median responses significantly greater than the median response to natural images ($p < 0.02$, one-sided permutation test, $n = 10$, asterisks) except one set of maximizing synthesized stimuli (bottom row, V4 neuron 5, red dots, $p = 0.922$, one-sided permutation test, $n = 10$). The response-maximizing natural images were reproduced from Adobe Stock. **b**. Real V4 responses to the artificial dot stimuli that varied in dot location (left), dot size (middle) and number of dots (right). Same format as in Fig. 4b and Extended Data Fig. 8b. Dot locations were subsampled to 28 × 28 locations to limit the number of images. Error bars in 'vary dot number' denote 1 s.e.m. across 10 different images, where each image had the same number of dots but in randomly-chosen, non-overlapping locations within 25 pixels of the preferred dot location. We found that these V4 neurons had preferred dot locations (left column), preferred dot sizes (middle column), and preferred numbers of dots (right column), consistent with these V4 neurons being dot detectors. Thus, these results provide strong evidence for the presence of dot detectors in V4. We observed diverse selectivity to dot size, including V4 neurons selective to the tiniest dots (neurons 1, 2, and 5) and small dots (neurons 3 and 4). Similarly, we observed selectivity to one dot (neurons 2 and 4), 2 dots (neuron 3), and 3 or more dots (neurons 1 and 5). Thus, even within the class of dot detectors, there appears to be large diversity in stimulus preferences.



Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | Explaining a dot-detecting compact model's selectivity for dot number. To understand the inner computations of a compact model, we chose to investigate a particular compact model that resembled a dot detector (Fig. 4). We focused on the model's selectivity to dot size (Fig. 4b) and uncovered a simple computation by isolating the filters that contributed to dot size selectivity (Fig. 4c–h). However, we suspected this compact model was also selective to the number of dots, as its preferred stimuli typically had two to five dots in the image (Fig. 4a). In addition, it was unclear to what extent dots fully explained the model's responses, which likely also depend on other complicated feature processing (for example, the texture surrounding the dot in the response-maximizing synthesized images, Fig. 4a). Here, we investigate this model's preference for dots and focus on how dot number selectivity may arise in a compact model. **a**. Diagram of the compact model that resembles a dot detector. We expected to see dot-like filters (that is, center-surround filters with middle excitation surrounded by inhibition or vice versa) in layers 1-3 but found none. This led us to identify corner-curvature and large-edge detecting filters that ultimately contributed to dot size selectivity (Fig. 4c–h). **b**. Besides varying dot size, we also varied the location of a small dot with a radius of 5 pixels (left, 'vary dot location') and dot number (right, 'vary dot number'). For dot number, we chose the location of the dots in a greedy manner by choosing the location of the next dot that maximized the model's response. This compact model (same as in **a** and Fig. 4) preferred two to three dots to the center right (with dot radii of 5 pixels). **c**. In the same manner as we identified filters contributing to dot size selectivity (Fig. 4c,d), we ablated each filter and measured the model's dot number invariance. A dot number invariance of 1 indicates that after ablating a filter, the model is no longer able to detect different numbers of dots; a dot number invariance of 0 indicates no change in the model's output after filter ablation (that is, dot number selectivity remains intact). Similar to dot size selectivity (Fig. 4e), we found that the most contributing filters resided in layers 4 and 5, consistent with the idea that a compact model's specialization occurs in its consolidation step. **d**. To better understand the chosen compact model's responses to dots versus other image features, we fed in images from different stimulus classes, including gratings, plaids, dots, and combinations of the three. Background colors were sampled from colors found in natural images. We found that punctate, black dots yielded the largest responses ('black dots') but that the response range was well above that for natural images (compare 'black dots' and 'natural'), suggesting that black dots alone did not capture the full response profile of the model. Considering dots that varied in size and color ('dots') as well as different combinations of gratings and plaids (with different orientations and spatial frequencies) better matched the response distribution of natural images. For the combination of dots, gratings, and plaids, each feature (dots with different sizes and colors, black dots, gratings, and plaids) was randomly added with a chance of 0.5. Natural greyscale images ('natural greyscale') tended to drive responses less than those to natural images ('natural'), suggesting the compact model encodes color. The natural greyscale images were adapted from Adobe Stock. The natural images were reproduced from Adobe Stock. **e**. To assess the importance of dots to the chosen compact model, we used the compact model's responses to each stimulus class in **d** as training data to train a "student" DNN model (same architecture as the distilled model with 5 layers, 100 filters per layer, see Methods) to match as close as possible the original compact model (the "teacher" model). We evaluated each student model's prediction performance (noise-corrected R^2) on held-out real V4 responses to natural images for the V4 neuron corresponding to the chosen compact model. For each student model, we trained on the same amount of data (12 million images), where each image was either artificially generated (dots, gratings, plaids, …) or from our image dataset (greyscale natural and natural images). Each bar denotes the noise-corrected R^2 for this particular V4 neuron ($n = 1$). We found that training separately on gratings, plaids, gratings and plaids, or black dots led to poor prediction performance compared to the original (orange bar). By increasing the complexity of the artificial stimuli to include dots of different sizes and colors ('dots'), as well as to add gratings and plaids to the background ('dots+gratings', 'dots+plaids', 'dots+grats+plaids'), improved performance, especially for dots and plaids. The prediction performance for training on dots and plaids was only 90% of the original performance ('dots+plaids' versus 'original'). This suggests that including stimuli that strongly drive the compact model (dots) as well as stimuli that weakly drive the compact model (plaids) are both important to capture the response distribution of natural images (compare response distributions in **d**). Finally, training on greyscale natural images had worse performance than training on colorful natural images (grey bar versus black bar), indicating the importance of color for the compact model. We note that the noise-corrected $R^2 = 0.4$ for this neuron suggests that the neuron itself uses computations not captured by the compact model although the stimulus preferences are likely the same (Fig. 2). **f**. To understand the specific computations of dot number selectivity, we fed in three input images with different numbers of dots ('input image'). We then observed the resulting activity and filter weights for layer 5 and the spatial readout layer. We noticed that the input to layer 5 ('layer 5 input') appeared to detect the presence of a dot at any given location (matching our intuition for identifying a single dot in Fig. 4f–h). The layer 5 filters (chosen as those with the largest dot number invariance in **c**) appeared to extract specific patterns of the detected dots, with sparse weights that have large magnitudes on the perimeters (for example, the first filter in layer 5 has large excitatory weights on the four corners). This spacing of the weights enables the filters to detect multiple dots spaced apart while not responding to dots too close together (as these would not align with the filter weights). After convolution with layer 5 filters, some filters formed large excitatory regions around the dots whereas others formed smaller regions of inhibition ('layer 5 conv. output'). After a linear combination of these filters (by multiplying with readout matrix W_{readout}) and the ReLU activation function, we found that most of the dot regions were extinguished with some filters activated by the boundary (or shell) of the dot region ('spatial readout layer input'). This activity was then excited or suppressed by spatial readout filters that act as spatial receptive fields. After taking a linear combination over filters ('sum over filters'), we observed that a single small dot had both excitation and inhibition (top activity map), while three dots led to large activity around the region of the dots (middle activity map). Many dots led to little to no activity (bottom activity map). Summing over spatial information across pixels recreated the selectivity to dot number (rightmost plot). **g**. We illustrate the computational motifs in **f** with a conceptual diagram. The key concept is that the model identifies a region of dots with filters that have sparse weights (encouraging distance between dots), extracts the shell around this region (by creating a larger excitatory region and a smaller inhibitory region), and queries the size of this shell. Too small a region (a single dot) leads to weak activity and inhibition (rightmost, top activity map). Too large a region leads to weak excitation, as the shell of the region is outside the spatial receptive field. Only an appropriately sized region (that is, a specific number of dots) will fit within the spatial receptive field, leading to strong excitation. These results suggest two key computations for dot number selectivity: 1) filters with sparse weights that identify spaced-apart dots and 2) extracting "regions" of interest and then identifying the boundary of these regions. If the region of interest is too large, the region's boundary will be outside the spatial receptive field, yielding a small response. These computations may comprise reoccurring themes in the visual cortex for other visual features.



Extended Data Fig. 9 | Other dot detecting compact models share similar computations for dot size selectivity. We chose to investigate a single compact model that strongly resembled a dot detector (Fig. 4), but we found other compact models whose preferred stimuli contained dots (Fig. 2a). We wondered whether these other dot-detecting compact models had similar computations for dot size selectivity as the compact model we investigated. We chose two other compact models, performed the same analyses as for our chosen compact model (Fig. 4), and found that all three models used similar computations. This suggests that these compact models converged to the same solutions despite different random initializations and V4 neurons. **a**. Response-maximizing natural and synthesized images for a second compact model. The response-maximizing natural images were reproduced from Adobe Stock. **b**. Stimulus tuning to artificial dot stimuli (same stimuli as in Fig. 4b and Extended Data Fig. 8b). This compact model preferred a dot to the bottom left with a dot size of 5 pixels and 3-5 dots. **c**. We computed the dot size invariance (DSI) for each filter by ablating that filter and comparing the model's outputs to its outputs for no ablation (see Methods, same procedure as in Fig. 4c,d). Filters with the largest DSIs (that is, filters that strongly contribute to dot size selectivity) were in the deeper layers (4,5, and spatial readout). **d**. We performed cumulative ablation—ablating all previously-chosen filters and choosing the next filter to ablate that led to the smallest increase in DSI (same procedure as in Fig. 4e)—to the filters in layer 3 and found roughly 10 filters together strongly contributed to dot size selectivity (rightmost dots). **e**. We investigated 6 filters that contributed to dot size selectivity; the other contributing filters appeared to serve redundant computations as the chosen filters. Much like the first chosen compact model (Fig. 4f), we found 4 excitatory filters that detected corners of dots ('layer 3 filters', top 4 filters with pink weights) and two inhibitory filters that detected large edges ('layer 3 filters', bottom 2 filters with green weights). In response to an image of a small dot, summing the output of these filters ('layer 3 conv. output') led to a large amount of overlap for the outputs of the excitatory filters but not for the inhibitory filters. The resulting large activity after layer 4 ('layer 4 output'), once summed across pixels, would lead to a dot being detected. We note that the layer 4 filter ('layer 4 filter') has sparse weights that likely promotes the detection of multiple dots, similar to what we found for the first chosen compact model (Extended Data Fig. 8). **f**. Same analysis of internal activations of the compact model as in **e** except for an input image of a large dot. Here, summing across filter outputs causes overlap between excitatory and inhibitory activity, cancelling each other out. This leads to weak levels of output activity ('layer 4 output') that results in no dot being detected. Comparing the results of **a**–**f** with the results of the first chosen compact model (Fig. 4), we conclude that both compact models use almost identical computations for dot size selectivity. **g**. Same analyses as in **a**–**f** except for a different compact model. We found that this third compact model's computations were largely similar to those of the other two dot-detecting compact models. The response-maximizing natural images were reproduced from Adobe Stock.

nature portfolio

Corresponding author(s): Benjamin R. Cowley

Last updated by author(s): Nov 13, 2025

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Stimulus generation was performed with Psychtoolbox (v3.0.14, www.psychtoolbox.org). Infrared eye tracking was EyeLink 1000 (SR Research, Ottawa, Ontario). Neural data acquisition was done using the Grapevine system (Ripple, Salt Lake City, UT). All spike sorting was performed using custom software written in Matlab (version R2021b, Mathworks, Inc., Natick, MA) available at https://github.com/smithlabvision/spikesort.
Data analysis	Code for all analyses was custom written in Python (v3.6.8). Custom-written code related to the deep neural network models used Keras (v2.2.4) on top of TensorFlow (v1.14.0). Some of the task-driven DNNs were only available in PyTorch (v2.2.1+cu121). Model weights, stimulus images, processed data, and code are available at https://github.com/cowleygroup/V4_compact_models.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Raw data are available at https://doi.org/10.1184/R1/30500090. Processed data and stimulus images are available at https://github.com/cowleygroup/V4_compact_models.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	N/A
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We analyzed data from three rhesus monkeys. No statistical methods were used to pre-determine sample size, but our sample size was consistent with the standard for non-human primate electrophysiological studies.
Data exclusions	We excluded neurons whose waveforms were not stably recorded throughout a session (see Online Methods for exclusion criteria) and images with less than 2 repeats.
Replication	The main results were reliably reproduced in three subjects, as described in the manuscript and Extended Data Figures.
Randomization	Randomization was not relevant to this study, because there was no data grouping.
Blinding	Blinding was not relevant to this study, because there was no data grouping.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Three adult male animals (Macaca mulatta) were used for this study. Their ages were 13, 12, and 7 years for WE, PE, and RA, respectively.
Wild animals	No wild animals were used in this study.
Reporting on sex	All animals were male.
Field-collected samples	No field-collected samples were used in this study.
Ethics oversight	Experimental procedures were approved by the Institutional Animal Care and Use Committee of Carnegie Mellon University and were performed in accordance with the United States National Research Council's Guide for the Care and Use of Laboratory Animals.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A